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Enhanced indexation: can volatility timing improve portfolio performance?

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This paper proposes a volatility timing-based enhanced indexation strategy that minimizes the variance of differences in returns between a tracking portfolio and a volatility timing benchmark. The volatility timing benchmark is formulated by embedding a dynamic volatility timing factor into the original benchmark index. We establish and solve the volatility timing-based enhanced indexation model, and find that the optimal portfolio consists of the global minimum variance portfolio and a mimicking portfolio that captures the volatility timing benchmark returns. Using data from the US and Chinese stock markets, we show that the optimal portfolio outperforms the benchmark index and seven other benchmark portfolios in terms of excess returns and risk-adjusted returns. The superior performance of the proposed strategy can be attributed to its enhanced upside participation and downside participation achieved through volatility timing.

Keywords: Enhanced indexation; Volatility timing; Benchmark index; Risk-adjusted return

JEL classifications: G10, G11, G15, G17

1. Introduction

In recent decades, index-linked funds have experienced substantial growth and gained widespread popularity among investors (Anis *et al.* 2023, Huang *et al.* 2024). These funds can be classified into index tracking funds and enhanced indexation funds. The index tracking funds aim to replicate the returns of a given benchmark index, such as the S&P 500 (Penev *et al.* 2022). In contrast, the enhanced indexation funds aim to outperform the benchmark index's returns while tracking its trends (Huang *et al.* 2022a). The enhanced indexation funds have proven appealing to investors and experienced more rapid growth than the index tracking funds. This is due to their ability to deliver excess returns and track index trends (Yang and Huang 2022, Dai and Li 2024).

This paper investigates the enhanced indexation problem via a novel volatility timing approach. The enhanced indexation problem is typically constructed as a dual-objective optimization problem, seeking to minimize tracking errors while maximizing excess returns (Li *et al.* 2022). In the mean-variance framework, prior literature has established the mean-tracking error variance enhanced indexation model, defining tracking error as the variance of the differences in

returns between the tracking portfolio and the benchmark index, and excess return as the difference between their expected returns (e.g. Roll 1992, Yang and Huang 2022, Ling *et al.* 2023). Other researchers use different measures for tracking error, such as mean square error (Beasley *et al.* 2003, Huang *et al.* 2018, Li *et al.* 2022), Weighted Conditional Value-at-Risk (Goel *et al.* 2018, Guastaroba *et al.* 2020), and lower partial moment (Guastaroba *et al.* 2016, Huang *et al.* 2022a, 2022b), among others.

Despite the use of various tracking error metrics, the tracking portfolio maintains constant exposure to the benchmark index in these tracking error metrics (Jiang *et al.* 2021). This makes the tracking portfolio sensitive to fluctuations in the benchmark index returns. Hence, when the benchmark index falls, especially in volatile times, investors may experience large losses (Wang *et al.* 2012). Huang *et al.* (2018, 2022b) also point out that the tracking portfolio is likely to face jump risks during index recessions, emphasizing the issue of maintaining constant exposure to a declining benchmark index. Therefore, it is necessary to consider the risks related to the benchmark index itself, such as its conditional volatility (Ahmed *et al.* 2024), when making index investing decisions.

To address this concern, an intuitive approach is to dynamically adjust the tracking portfolio's exposure to the

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benchmark index, based on changes in the benchmark index's conditional volatility. This strategy, known as volatility timing, can be considered a risk-based strategy, without relying on any measures of expected returns. Fleming *et al.* (2001, 2003) initially demonstrate the economic value of volatility timing within the mean-variance framework. Subsequent studies show that volatility timing can improve the Sharpe ratio or alpha of different investment strategies, such as active volatility timing strategies (Kirby and Ostdiek 2012), portfolio insurance strategies (Albeverio *et al.* 2013, Zieling *et al.* 2014), and factor investing strategies (Moreira and Muir 2017, Xiong *et al.* 2022). However, the exploration of volatility timing in enhanced indexation remains a valuable area that has yet to be investigated.

This paper proposes a volatility timing-based enhanced indexation strategy that minimizes the variance of the differences in returns between the tracking portfolio and a benchmark index adjusted by a dynamic volatility timing factor. Inspired by Barroso and Santa-Clara (2015) and Kirby and Ostdiek (2012), the volatility timing factor is formulated as a power function of the inverse of the conditional volatility of the benchmark index returns, enabling the tracking portfolio to adjust exposures based on the changes of the benchmark index's conditional volatility. In addition, unlike traditional enhanced indexation models that require estimating expected returns of both risky assets and the benchmark index (de Paulo *et al.* 2016), the proposed strategy relies solely on estimating their covariance and variance, mitigating the impact of estimation errors of expected returns (Jiang *et al.* 2021).

We establish and solve the volatility timing-based enhanced indexation model. We find that the volatility timing-based enhanced index portfolio (VEIP) comprises the global minimum variance portfolio (GMVP) and a volatility timing benchmark (VTB)-mimicking portfolio that captures the volatility timing benchmark's returns. We decompose the VTB-mimicking portfolio into a residual mimicking portfolio and a traditional mimicking portfolio that closely captures benchmark index returns. We find that, given a residual mimicking portfolio, the proportions of the traditional mimicking portfolio and the GMVP in the VEIP can vary to capture benchmark index returns or mitigate portfolio risks, depending on the beta of the volatility timing benchmark with respect to the benchmark index. Moreover, the beta of the VEIP is an increasing function of the beta of the volatility timing benchmark.

Using data from the US and Chinese stock markets, we demonstrate the significant economic value of the proposed strategy for investors in enhanced indexation. We first show that the volatility timing benchmark exhibits superior in-sample average returns and risk-adjusted returns compared to the benchmark index, supporting that the VEIP can achieve superior performance by tracking the volatility timing benchmark. Using a rolling window approach, we then demonstrate that the VEIP outperforms the benchmark index and seven other benchmark portfolios in terms of out-of-sample excess return, Jensen's alpha, Sharpe ratio, and omega ratio, regardless of whether short sales are allowed. Despite higher transaction costs due to the incorporation of volatility timing, the VEIP still achieves superior out-of-sample net excess returns and net Sharpe ratios after accounting for transaction costs. These performances can be attributed to the VEIP's superior

upside participation and downside participation achieved through volatility timing.

The contributions of this paper are as follows. First, this paper extends prior literature on indexation strategies by proposing a novel volatility timing-based enhanced indexation strategy. Prior studies develop enhanced indexation strategies based on balancing tracking errors and excess returns within the risk-return framework (e.g. Li and Bao 2014, Li *et al.* 2022, Huang *et al.* 2022a, 2022b). Other research uses methods, like second-order stochastic dominance (Roman *et al.* 2013), quantile regression (Mezali and Beasley 2013), integration of ESG information of assets (Li *et al.* 2022, Ling *et al.* 2023), and deep learning methods (Dai and Li 2024), to develop enhanced indexation strategies. Distinct from these studies, our strategy integrates the conditional volatility of the benchmark index into index tracking. This enables the tracking portfolio to adjust exposure, based on changes in the benchmark index's risks, achieving market-beating performance.

Second, this paper contributes to the literature on volatility timing by validating its economic value in index investing. Prior studies confirm the benefits of volatility timing in improving the Sharpe ratio of mean-variance strategies (Fleming *et al.* 2001, 2003, Kirby and Ostdiek 2012) and portfolio insurance strategies (Zieling *et al.* 2014), and its capabilities of enhancing alpha of common asset pricing factors (e.g. Moreira and Muir 2017, Wang and Yan 2021, Xiong *et al.* 2022). This paper extends the literature by demonstrating that volatility timing can deliver superior excess returns and risk-adjusted returns for investors in enhanced indexation. Further analysis shows that the superior performances are attributed to the improved market participation achieved through volatility timing. These results highlight the significant economic value of volatility timing in index investing.

Finally, this paper contributes to the literature on unconditional portfolio selection with conditioning information by incorporating the benchmark index's conditional volatility into the unconditional tracking model. Previous studies in this area use conditioning information in active or tracking portfolios to estimate parameters such as expected returns and covariance matrix of risky assets (e.g. Ferson and Siegel 2001, Ferson *et al.* 2006, Chiang 2015, Pereira 2021, Siegel 2021). In contrast, we incorporate into the tracking models the conditional risk information from the benchmark index, rather than the conditional expected returns and covariance matrix of risky assets. This adds a new dimension to incorporating conditioning information in unconditional tracking models.

The paper is organized as follows. Section 2 proposes the volatility timing-based enhanced indexation strategy. Section 3 provides the empirical design and datasets. Section 4 conducts empirical analysis. Section 5 concludes the paper.

2. Volatility timing-based enhanced indexation strategy

2.1. The model

Suppose investors' investment universe contains n candidate risky assets with a column return vector $\mathbf{r}$. A tracking portfolio is a vector $\mathbf{w} = (w_1, w_2, \dots, w_n)^T$, where w_i is the weight of

the i -th risky asset, and T is the transpose operation. Thus, the return of the tracking portfolio is $r_P = \mathbf{w}^T \mathbf{r}$. When the tracking portfolio tracks a benchmark index with a random return r_I , it inherently maintains a constant exposure to the benchmark index in tracking error metrics (Jiang *et al.* 2021).[†] This constant exposure, however, makes the tracking portfolio vulnerable to potential losses during periods in which the benchmark index declines (Wang *et al.* 2012, Huang *et al.* 2018).

To mitigate this issue, we introduce a dynamic volatility timing factor that adjusts the tracking portfolio's exposure based on changes in the conditional volatility of the benchmark index returns. Specifically, at time t , we multiply the benchmark index return $r_{I,t}$ by the dynamic volatility timing factor ϕ_t , such that the tracking portfolio actually tracks a volatility timing benchmark, denoted as $\phi_t r_{I,t}$. The tracking error at time t is then defined as the difference between the tracking portfolio return $r_{P,t}$ and the volatility timing benchmark return $\phi_t r_{I,t}$,

$$TE_t = r_{P,t} - \phi_t r_{I,t} = \mathbf{w}^T \mathbf{r}_t - \phi_t r_{I,t}. \quad (1)$$

Inspired by Kirby and Ost diek (2012) and Barroso and Santa-Clara (2015), the volatility timing factor is proposed as follows:

$$\phi_t = \left(\frac{\rho}{\sigma_{c,t}} \right)^\eta, \quad (2)$$

where $\sigma_{c,t}$ represents the conditional volatility of the benchmark index returns at time t . The parameters ρ ($\rho > 0$) and η ($\eta > 0$) are the target volatility and timing intensity, respectively. These parameters distinguish between high and low conditional volatility and determine how aggressively or conservatively the volatility timing factor responds to changes in the conditional volatility.

The motivations for constructing the volatility timing factor are intuitive. First, the volatility timing factor varies inversely with the conditional volatility, allowing the tracking portfolio to reduce exposure to potentially mitigate losses in highly risky periods (Dierkes and Krupski 2022) and even capture returns in less risky periods. This approach is supported by the generally negative correlation between the conditional volatility and concurrent benchmark index return, as noted by Busse (1999). Second, the volatility timing benchmark is an in-sample portfolio with leverage that can mitigate the impact of parameter estimation errors and capture returns with leverage. Furthermore, the parameters ρ and η allow investors to adjust the volatility timing factor to meet their risk tolerance and timing preferences (Kirby and Ost diek 2012, Barroso and Santa-Clara 2015). For brevity, we omit the time subscript t in subsequent discussions.

Further, the unconditional tracking error variance (TEV) is formulated as:

$$TEV = \mathbf{w}^T \mathbf{V} \mathbf{w} - 2\mathbf{w}^T \text{cov}(\mathbf{r}, \phi r_I) + \sigma_{VT}^2, \quad (3)$$

where $\mathbf{V}$ denotes the non-singular covariance matrix of risky asset returns. $\text{cov}(\mathbf{r}, \phi r_I)$ is a column vector representing the

covariance between each risky asset's return and the volatility timing benchmark return. σ_{VT}^2 is the variance of the volatility timing benchmark returns.

Given the independence between σ_{VT}^2 and $\mathbf{w}$, the proposed volatility timing-based enhanced indexation strategy that aims to minimize the first two terms of the TEV is given as:

$$\begin{cases} \min_{\mathbf{w}} \mathbf{w}^T \mathbf{V} \mathbf{w} - 2\mathbf{w}^T \text{cov}(\mathbf{r}, \phi r_I) \\ \text{s.t. } \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (4)$$

where $\mathbf{w}^T \mathbf{1} = 1$ is the weight budget constraint and $\mathbf{1}$ is a column vector of ones.

Model (4) embeds the conditional volatility of the benchmark index, distinguishing it from unconditional mean-TEV models that rely on excess return measures (Roll 1992, Chiang 2015). Furthermore, it avoids estimating expected returns for the risky assets and the benchmark index, mitigating the impact of parameter estimation errors on portfolio weights (Jiang *et al.* 2021).

2.2. Optimal solution

PROPOSITION 1 When short sales are allowed, the optimal solution $\mathbf{w}^*$ of the Model (4) ($n \geq 2$) is given by:

$$\mathbf{w}^* = a\mathbf{w}_{VT} + (1-a)\mathbf{w}_G, \quad (5)$$

where $a = \mathbf{1}^T \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi r_I)$, $b = \mathbf{1}^T \mathbf{V}^{-1} \mathbf{1}$, $\mathbf{w}_{VT} = \frac{1}{a} \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi r_I)$, and $\mathbf{w}_G = \frac{1}{b} \mathbf{V}^{-1} \mathbf{1}$.

Proof See Appendix A.

In Equation (5), the VEIP is a combination of $\mathbf{w}_{VT}$ and the GMVP, $\mathbf{w}_G$, derived from the mean-variance model. According to Chiang (2015), $\mathbf{w}_G$ serves to reduce the variance of the optimal portfolio returns. To understand the construction of $\mathbf{w}_{VT}$, we perform a linear regression of the volatility timing benchmark returns on individual assets' returns:

$$\phi r_I = \alpha_{VT} + \boldsymbol{\gamma}_{VT}^T \mathbf{r} + \varepsilon_{VT}, \quad (6)$$

where α_{VT} is an intercept, $\boldsymbol{\gamma}_{VT} = (\gamma_1, \gamma_2, \dots, \gamma_n)^T$ is the regression coefficient vector, and ε_{VT} is the error term. We assume that $E[\varepsilon_{VT}] = 0$ and $\text{cov}(\mathbf{r}, \varepsilon_{VT}) = 0$. $\alpha_{VT} + \varepsilon_{VT}$ and $\boldsymbol{\gamma}_{VT}^T \mathbf{r}$ are the parts of the volatility timing benchmark returns that the n risky asset returns cannot and can explain, respectively. ■

Substituting Equation (6) into the expression of $\mathbf{w}_{VT}$, we have:

$$\begin{aligned} \mathbf{w}_{VT} &= \frac{\mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi r_I)}{a} = \frac{\mathbf{V}^{-1} \text{cov}(\mathbf{r}, \alpha_{VT} + \boldsymbol{\gamma}_{VT}^T \mathbf{r} + \varepsilon_{VT})}{\mathbf{1}^T \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \alpha_{VT} + \boldsymbol{\gamma}_{VT}^T \mathbf{r} + \varepsilon_{VT})} \\ &= \frac{\boldsymbol{\gamma}_{VT}}{\mathbf{1}^T \boldsymbol{\gamma}_{VT}}. \end{aligned} \quad (7)$$

In Equation (7), $\mathbf{w}_{VT} = \boldsymbol{\gamma}_{VT} / \mathbf{1}^T \boldsymbol{\gamma}_{VT}$ is the normalized regression coefficient vector. Hence, $\mathbf{w}_{VT}$ can capture the volatility timing benchmark returns using the risky asset returns as closely as possible, and thus we define it as the VTb-mimicking portfolio.

[†] Throughout the paper, the risk-free rate is excluded from the returns on both the risky assets and index.

2.3. Role of volatility timing factor

We further explore how the volatility timing factor impacts the VTB-mimicking portfolio as well as the VEIP. Utilizing the CAPM model,[†] we have:

$$\mathbf{r} = \boldsymbol{\alpha} + \boldsymbol{\beta}r_I + \boldsymbol{\varepsilon}, \quad (8)$$

where $\boldsymbol{\alpha}$ is the intercept vector, $\boldsymbol{\beta} = \text{cov}(\mathbf{r}, r_I)/\sigma_I^2$ is the vector of beta coefficients for the risky assets relative to the benchmark index, and $\boldsymbol{\varepsilon}$ is the error vector. $\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$ and $\boldsymbol{\beta}r_I$ are the parts of the risky asset returns that the benchmark index cannot explain and can explain, respectively.

Substituting Equation (8) into the VEIP, we have:

$$\begin{aligned} \mathbf{w}^* &= a\mathbf{w}_{VT} + (1-a)\mathbf{w}_G \\ &= \mathbf{V}^{-1}\text{cov}\left(\boldsymbol{\alpha} + \left(\frac{\text{cov}(\mathbf{r}, r_I)}{\sigma_I^2}\right)r_I + \boldsymbol{\varepsilon}, \phi r_I\right) + (1-a)\mathbf{w}_G \\ &= \frac{\text{cov}(\phi r_I, r_I)}{\sigma_I^2}\mathbf{V}^{-1}\text{cov}(\mathbf{r}, r_I) + \mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I) \\ &\quad + (1-a)\mathbf{w}_G \\ &= \beta^{VT}c\mathbf{w}_M + d\mathbf{w}_\varepsilon + (1-\beta^{VT}c-d)\mathbf{w}_G \end{aligned} \quad (9)$$

where $\beta^{VT} = \text{cov}(\phi r_I, r_I)/\sigma_I^2$, $c = \mathbf{1}^T\mathbf{V}^{-1}\text{cov}(\mathbf{r}, r_I)$, $\mathbf{w}_M = \frac{1}{c}\mathbf{V}^{-1}\text{cov}(\mathbf{r}, r_I)$, $d = \mathbf{1}^T\mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I)$, and $\mathbf{w}_\varepsilon = \frac{1}{d}\mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I)$.

Following Equation (7), $\mathbf{w}_M$ can be considered as the traditional mimicking portfolio, aiming to capture the benchmark index returns with risky asset returns. Hence, $\beta^{VT}c\mathbf{w}_M$ captures the volatility timing benchmark return with risky asset returns, as β^{VT} is the beta of the volatility timing benchmark relative to the benchmark index. Moreover, $\mathbf{w}_\varepsilon = \mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I)/d$ aims to capture the volatility timing benchmark return using the error vector. In this context, given a $\mathbf{w}_\varepsilon$, β^{VT} determines the proportion of the traditional mimicking portfolio and GMVP in the VEIP.

To explore the effect of β^{VT} , we reformulate it as follows:

$$\begin{aligned} \beta^{VT} &= \frac{\text{cov}(\phi r_I, r_I)}{\sigma_I^2} = \frac{E(\phi r_I^2) - E(\phi r_I)E(r_I)}{\sigma_I^2} \\ &= E(\phi) + \frac{\text{cov}(\phi, r_I^2)}{\sigma_I^2} - \frac{\text{cov}(\phi, r_I)E(r_I)}{\sigma_I^2}. \end{aligned} \quad (10)$$

In Equation (10), $E(\phi)$ is a constant for given ρ and η . Hence, β^{VT} is influenced by two key components: $\text{cov}(\phi, r_I^2)/\sigma_I^2$ and $-\text{cov}(\phi, r_I)E(r_I)/\sigma_I^2$, where $\text{cov}(\phi, r_I^2)/\sigma_I^2$ depends on the covariance between the volatility timing factor and the concurrently realized volatility of the benchmark index, and $-\text{cov}(\phi, r_I)E(r_I)/\sigma_I^2$ depends on the covariance between the volatility timing factor and the concurrent benchmark index return. Given $\phi = (\rho/\sigma_c)^\eta$, and the conditional volatility positively correlates with the concurrent realized volatility (Wang and Yan 2021), $\text{cov}(\phi, r_I^2)/\sigma_I^2$ is negative. Consequently, the VEIP increases the proportion of the GMVP component, reducing portfolio risks.

Furthermore, given the conditional volatility generally negatively correlates with the concurrent benchmark index return (Busse 1999), $-\text{cov}(\phi, r_I)E(r_I)/\sigma_I^2$ is negative. Consequently, the VEIP also increases the proportion of the GMVP component, thereby mitigating portfolio risks. Sometimes, when this negative correlation becomes positive, the VEIP increases the proportion of the traditional mimicking portfolio to capture the benchmark index returns. Collectively, these results indicate that the VEIP can adjust the proportions of the traditional mimicking portfolio and GMVP component to capture benchmark index returns or mitigate portfolio risks, depending on the changes of β^{VT} .

2.4. Beta of the VEIP

We further analyze the beta of the VEIP relative to the benchmark index, which reflects the degree of the unconditional risk exposure of the VEIP to the benchmark index and measures the degree to which the VEIP captures the benchmark index returns. Based on the definition of the beta, we have:

$$\beta_P = \frac{\text{cov}(\mathbf{w}^{*T}\mathbf{r}, r_I)}{\sigma_I^2} = \frac{\beta^{VT}(bc - d^2) + (be - cd) + d}{b\sigma_I^2}, \quad (11)$$

where $c = \text{cov}(\mathbf{r}, r_I)^T\mathbf{V}^{-1}\text{cov}(\mathbf{r}, r_I)$, $e = \text{cov}(\mathbf{r}, r_I)^T\mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I)$, and $d = \mathbf{1}^T\mathbf{V}^{-1}\text{cov}(\boldsymbol{\varepsilon}, \phi r_I)$.

Given $bc - d^2 > 0$ and $b > 0$, Equation (11) indicates that β_P is an increasing function of β^{VT} .[‡] Thus, as β^{VT} rises, the VEIP improves its exposure to the benchmark index, capturing a larger proportion of the benchmark index returns. Conversely, when β^{VT} decreases, the VEIP reduces its exposure, capturing a smaller fraction of the benchmark index returns. Consequently, the VEIP can adjust its exposure to the benchmark index in response to changes in β^{VT} , thereby adapting its aggressiveness or conservatism to capture benchmark index returns.

3. Empirical design and datasets

3.1. Benchmark portfolios

To assess the VEIP's effectiveness, we compare its out-of-sample performance with that of eight benchmark portfolios. These benchmark portfolios include the benchmark index, the traditional index tracking portfolio (ITP) (Rudolf et al. 1999), the factor tracking portfolio (FTP) (Jiang et al. 2021), the traditional enhanced index tracking portfolio (EIP) (de Paulo et al. 2016), the equally weighted index (EWI), the equally weighted portfolio (EWP) (DeMiguel et al. 2009), the GMVP (Jiang et al. 2019), and the volatility timing portfolio (VTP) (Kirby and Ostdiek 2012).

In all these portfolios considered, the VEIP and ITP/FTP differ in whether incorporate volatility timing. Hence, a comparison of them can reveal the effectiveness of volatility timing in enhanced indexation. The EWI and EWP are used as benchmarks because they perform well and do not require

[†] The risk-free rate is excluded from the returns on both the risky assets and index.

[‡] $(d\text{cov}(\mathbf{r}, r_I) - c\mathbf{1})^T\mathbf{V}^{-1}(d\text{cov}(\mathbf{r}, r_I) - c\mathbf{1}) = c(bc - d^2) > 0$, where $\mathbf{V}^{-1}$ is positive definite.

Table 1. List of the examined portfolios.

#	The examined portfolios	Abbreviation	Short sales	Literature
Panel A: Volatility timing-based enhanced indexation portfolio				
1	Volatility timing-based enhanced indexation portfolio	VEIP	Y&N	—
Panel B: Benchmark portfolios				
2	Benchmark index	—	N	—
3	Equally weighted index	EWI	N	—
4	Traditional index tracking portfolio	ITP	Y&N	Rudolf <i>et al.</i> (1999)
5	Traditional enhanced indexation portfolio	EIP	Y&N	Beasley <i>et al.</i> (2003)
6	Factor tracking portfolio	FTP	Y&N	Jiang <i>et al.</i> (2021)
7	Volatility timing portfolio	VTP	N	Kirby and Ostdiek (2012)
8	Equally weighted portfolio	EWP	N	DeMiguel <i>et al.</i> (2009)
9	Global minimum variance portfolio	GMVP	Y&N	Jiang <i>et al.</i> (2019)

Notes: Y denotes the allowance of short sales, whereas N indicates the disallowance.

parameter estimation or optimization (DeMiguel *et al.* 2009). By including the GMVP, a part of the VEIP, we can reveal whether the VEIP as well as the VTB-mimicking portfolio captures the superior returns from the volatility timing benchmark using risky assets. In addition, comparing VEIP with EIP and VTP elucidates its advantages over these strategies.

Furthermore, we analyze the out-of-sample performance of the VEIP, ITP, FTP, EIP, and GMVP under two scenarios: those without and with short sales.[†] In the latter scenario, we cap the total negative portfolio weight at 30%, aligning with the 130/30 strategy (Clarke *et al.* 2008). Due to inherent constraints, other portfolios do not allow short sales. Table 1 summarizes the portfolios examined, and Appendix B presents detailed models of the benchmark portfolios.

3.2. Parameter estimation

For the portfolios being examined, aside from the benchmark index, EWI, and EWP, we need to estimate the expected returns and covariance matrix of risky assets, as well as the conditional volatility of the benchmark index. To this end, sample estimates are used for the expected returns and covariance matrix, and conditional volatility is estimated using daily benchmark index returns for each month (Barroso and Santa-Clara 2015):

$$\hat{\sigma}_{c,t} = \sqrt{\frac{252}{J-1} \sum_{j=1}^J (r_{dj,t} - \bar{r}_d)^2}, \quad (12)$$

where J denotes the number of daily returns for the benchmark index in month t , $r_{dj,t}$ is the j -th daily return in month t , and $\bar{r}_d$ is the average daily return in month t . Furthermore, the conditional volatility is annualized by multiplying by $\sqrt{252}$. Then, the annualized conditional volatility $\hat{\sigma}_{c,t}$ is combined with ρ and η to calculate the volatility timing factor ϕ_t , which adjusts the benchmark index return $r_{I,t}$ to yield the volatility timing benchmark return $\phi_t r_{I,t}$ in each month. Notably, the volatility timing benchmark is not subject to look-ahead bias because it only uses benchmark index returns prior to each decision-making point.

Using these estimates, we use a rolling window approach to assess the out-of-sample performance of each strategy. Specifically, for a given sample period of K months, we use the initial $L = 96$ months of data (where $L < K$) to estimate parameters and calculate volatility timing benchmark returns. These returns and estimates are then used to optimize portfolios, which are used to compute one-month out-of-sample returns for each strategy. This process is repeated monthly, advancing the estimation window until the end of the sample. Consequently, we obtain a series of optimal portfolios and out-of-sample returns for each strategy, allowing us to compute their performance metrics, including the excess return, Jensen's alpha, standard deviation, Sharpe ratio, Omega ratio, turnover, net excess return, and net Sharpe ratio. These indicators are defined in Appendix C.

3.3. Datasets

In empirical analysis, we use the S&P 500 and CSI 300 indexes, which have been used in prior research (Li *et al.* 2022, Tu and Li 2024). Bharjana *et al.* (2023) suggest that investors can use stratified index portfolios, such as sector exchange-traded funds (ETFs) or indexes, to create tracking portfolios. This simplifies the complex stock selection process, covers broad industries, and reduces tracking errors, idiosyncratic risks, and transaction costs. Therefore, we construct portfolios using the CSI 300's 10-sector indexes and the S&P 500's 11-sector indexes (excluding real estate due to missing data). The CSI 300 dataset covers a period from January 2003 to December 2022, and the S&P 500 dataset spans January 2005 to December 2022. Both datasets include monthly and daily linear index returns and monthly risky asset linear returns adjusted for risk-free rates. The data are obtained from the Wind database, except for the US risk-free rate sourced from Kenneth French's website.[‡] For detailed sector information, refer to Appendix D.

Figure 1 illustrates the cumulative returns for the S&P 500 and the CSI 300 indexes. We observe an upward trend for the S&P 500 index and significant volatility for the CSI 300 index.

[†] There are a small number of index-linked funds that allow short selling.

[‡] http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

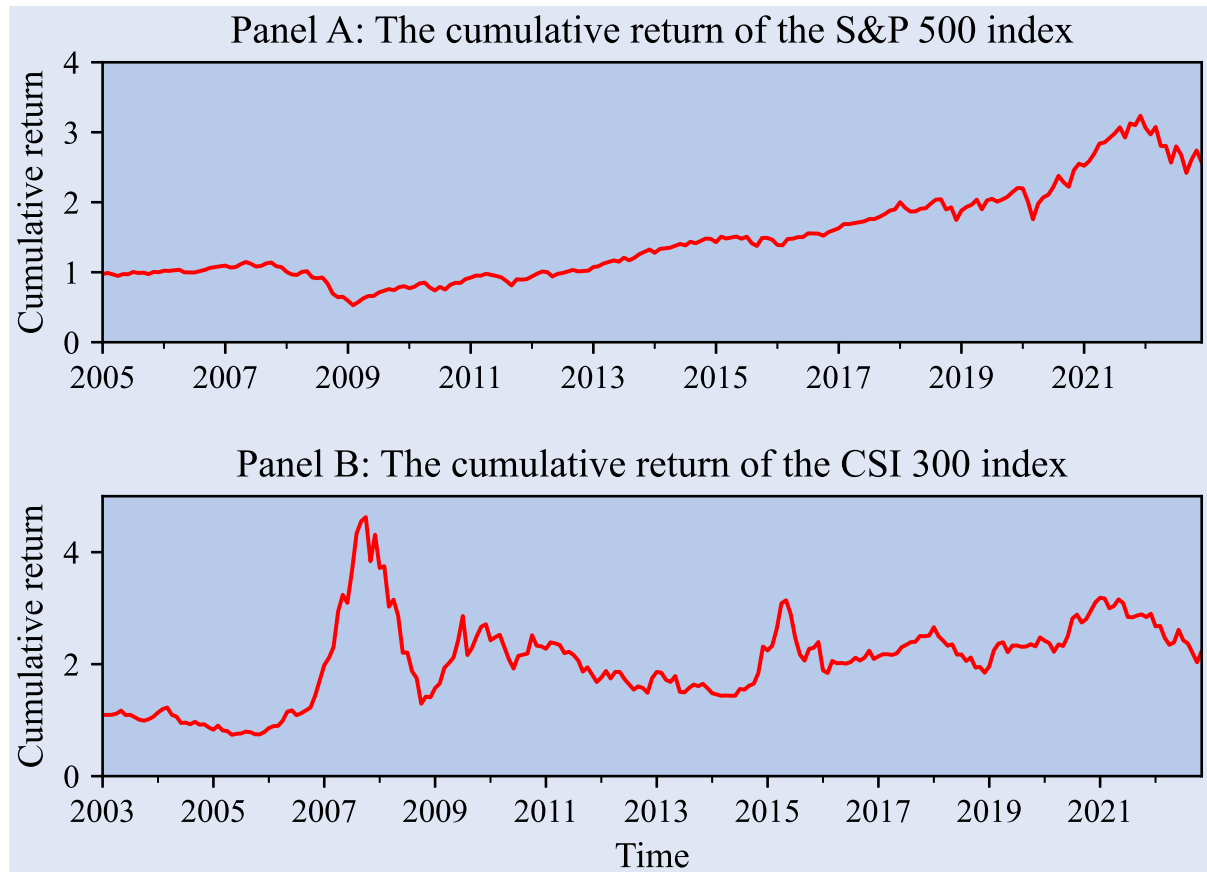


Figure 1. The cumulative returns of both the S&P 500 index and the CSI 300 index.

Table 2. Descriptive statistics of the risky assets.

Panel A: Descriptive statistics of the US ten sector indexes in the S&P 500 index										
	Condc	Consp	Helcr	Indst	Infct	Mteri	Comsr	Utilities	Fincl	Energy
AVR	0.078	0.062	0.082	0.065	0.112	0.064	0.015	0.050	0.033	0.071
STD	0.192	0.118	0.138	0.190	0.189	0.205	0.170	0.143	0.225	0.264
SR	0.407	0.524	0.593	0.344	0.589	0.313	0.086	0.353	0.145	0.267
Skew	-0.020	-0.422	-0.295	-0.472	-0.387	-0.276	-0.434	-0.685	-0.668	0.001
Kurt	4.705	3.737	3.468	4.900	3.246	4.333	3.200	3.763	5.846	6.282
J.B. stats	26.188	11.300	5.105	40.515	5.951	18.738	7.126	22.155	88.946	96.965
Panel B: Descriptive statistics of the ten sector indexes in the CSI 300 index										
	Energy	Mteri	Indst	Condc	Consp	Helcr	Fincl	Infct	Telcm	Utilities
AVR	0.065	0.088	0.068	0.098	0.184	0.128	0.096	0.057	0.076	0.052
STD	0.317	0.338	0.302	0.293	0.293	0.292	0.310	0.329	0.316	0.271
SR	0.204	0.259	0.226	0.336	0.630	0.439	0.309	0.172	0.240	0.193
Skew	0.098	0.096	0.236	0.009	0.040	0.469	0.502	0.101	0.350	0.687
Kurt	4.235	3.995	5.186	4.370	3.996	5.646	5.622	3.834	5.728	6.699
J.B. stats	15.626	10.273	50.007	18.761	9.976	78.824	78.843	7.358	79.303	155.709

Notes: This table reports the annualized average return (AVR), annualized standard deviation (STD), Sharpe ratio (SR), skewness (Skew), kurtosis (Kurt), and Jarque-Bera statistics (J. B. stats) of the risky assets.

Table 2 provides descriptive statistics for risky asset returns based on the two datasets. Based on the S&P 500 dataset (Panel A), the average returns of risky assets range from 1.5% to 11.2%, with Information Technology (Infct) exhibiting the highest average return of 11.2% and Communications Services (Comsr) the lowest at 1.5%. Consumer Staples (Consp) displays the lowest standard deviation of 11.8%, contrasting with Energy's highest standard deviation of 26.4%.

Health Care (Helcr) achieves the highest Sharpe ratio of 0.593, while Comsr shows a lower Sharpe ratio of 0.086. All assets in Panel A exhibit departures from the normal distribution, characterized by leptokurtosis and left tails. Based on the CSI 300 dataset (Panel B), risky assets demonstrate an average return range of 5.2% to 18.4%, with Consumer Staples (Consp) achieving the highest average return of 18.4% and Utilities the lowest at 5.2%. Utilities also exhibit the

Table 3. The performance of the benchmark indexes.

	AVR	STD	SR	Omega	Skew	Kurt
Panel A: In-sample performance of the volatility timing benchmark for the S&P 500 index ($\rho = 14\%$)						
S&P 500	0.064	0.153	0.421	1.379	-0.580	4.233
VTB ($\eta = 1$)	0.177	0.134	1.320	2.622	0.285	2.628
VTB ($\eta = 2$)	0.342	0.210	1.627	4.783	1.678	7.315
VTB ($\eta = 3$)	0.672	0.468	1.434	8.355	3.910	24.637
VTB ($\eta = 4$)	1.420	1.275	1.114	13.947	6.187	49.975
Panel B: In-sample performance of the volatility timing benchmark for the CSI 300 index ($\rho = 18\%$)						
CSI 300	0.079	0.276	0.285	1.253	-0.001	4.422
VTB ($\eta = 1$)	0.094	0.199	0.473	1.415	0.239	2.706
VTB ($\eta = 2$)	0.113	0.210	0.538	1.554	0.769	5.965
VTB ($\eta = 3$)	0.171	0.321	0.532	1.801	3.555	30.008
VTB ($\eta = 4$)	0.332	0.663	0.501	2.338	7.232	77.243

Notes: This table reports the in-sample annualized average return (AVR), annualized standard deviation (STD), Sharpe ratio (SR), Omega, skewness (Skew), and kurtosis (Kurt) for the benchmark index and the volatility timing benchmark. VTB is the volatility timing benchmark.

lowest standard deviation at 27.1%, while Materials (Mteri) display the highest at 33.8%. Consp and Infte exhibit the highest and lowest Sharpe ratios, respectively, at 0.630 and 0.172. Returns of all assets in Panel B do not follow the normal distribution, displaying leptokurtosis and slightly right tails.

4. Empirical results

This section evaluates the VEIP's effectiveness. We first assess the volatility timing benchmark's in-sample performance and then compare the VEIP's out-of-sample performance with eight benchmark portfolios. Next, we examine the VEIP's market participation and cumulative returns. Finally, we compare the VEIP's out-of-sample performance with that of the volatility timing benchmark and conclude with robustness tests.

When implementing the VEIP, choosing η and ρ is crucial. We follow Kirby and Ostdiek (2012) and choose $\eta \in \{1, 2, 3, 4\}$. For ρ , we use different values based on the market: 16%, 18%, and 20% for the CSI 300 index, and 12%, 14%, and 16% for the S&P 500 index. We choose the ρ values for two reasons. First, they are used in previous studies (Albeverio *et al.* 2013, Barroso and Santa-Clara 2015) and are acceptable to investors. Second, the CSI 300 index is more volatile, so a higher ρ for the CSI 300 index prevents the VEIP from being overly conservative. Due to space limits, we only show results with ρ being equal to 18% for the CSI 300 and 14% for the S&P 500. Additional results are in the robustness tests.

4.1. Performance of the benchmark indexes

Table 3 reports the in-sample performance of the volatility timing benchmark and the benchmark index based on both datasets. The volatility timing benchmark exhibits higher average returns than the benchmark index. For instance, based on the CSI 300 dataset, the volatility timing benchmark's average return ranges from 9.4% to 33.2%, while the benchmark index's average return is only 7.9%. This is due to three

reasons. First, the volatility timing factor can mitigate losses during high volatility periods and even capture returns in opposite situations due to the negative correlation between the index's conditional volatility and its contemporaneous returns (Busse 1999).[†] Second, the volatility timing benchmark can earn extra returns through leverage. Third, the volatility timing benchmark is an in-sample portfolio, reducing parameter estimation errors' impacts (Jiang *et al.* 2019). Moreover, the standard deviation of the volatility timing benchmark is higher than the benchmark index, particularly $\eta \in \{3, 4\}$. This is because the volatility timing factor becomes more aggressive, as η increases, resulting in higher standard deviations.

Furthermore, the volatility timing benchmark outperforms the benchmark index in terms of both the Sharpe ratio and omega ratio. Thus, for each unit of risk, the volatility timing benchmark has higher average returns than the benchmark index. Moreover, the Sharpe ratio of the volatility timing benchmark first rises with η and then decreases, and the omega ratio consistently increases with η . This suggests that the risk-adjusted return of the volatility timing benchmark rises with η , as the omega ratio measures risk-adjusted returns from a high moment perspective.

Overall, the results presented in Table 3 confirm the superior average returns and risk-adjusted returns of the volatility timing benchmark, supporting that the VEIP can achieve superior performance by tracking the volatility timing benchmark.

4.2. Out-of-sample performance

4.2.1. Out-of-sample performance without short sales.

Table 4 reports the out-of-sample performance of the portfolios considered without short sales based on both datasets. We analyze the performance from four aspects: portfolio returns, standard deviations, risk-adjusted returns, and transaction costs.

In terms of portfolio returns, including the excess return and Jensen's alpha, the VEIP outperforms the benchmark index,

[†] For the CSI 300 index and S&P 500 index over the entire sample, the correlations between the conditional volatility and current return are -0.15 and -0.44, respectively

Table 4. Out-of-sample performance of the examined portfolios without short sales.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0073	0.0011*	0.1420	0.7803	1.7844	0.0747	0.0028	0.7496
VEIP ($\eta = 2$)	0.0246	0.0021**	0.1510	0.8484*	1.8633	0.0939	0.0189	0.8112*
VEIP ($\eta = 3$)	0.0453	0.0032**	0.1663	0.8946**	1.9435	0.1150	0.0383	0.8520*
VEIP ($\eta = 4$)	0.0557	0.0040**	0.1727	0.9217**	1.9823	0.1022	0.0495	0.8852**
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	-0.0020	-0.0002	0.1492	0.6802	1.6726	0.0340	-0.0041	0.6671
FTP	-0.0020	-0.0002	0.1493	0.6798	1.6725	0.0342	-0.0041	0.6666
EIP	0.0173	0.0008	0.1667	0.7246	1.7052	0.1377	0.0089	0.6758
GMVP	-0.0201	0.0009	0.1253	0.6658	1.6269	0.0880	-0.0254	0.6241
EWP	-0.0114	-0.0006	0.1449	0.6358	1.6311	0.0320	-0.0133	0.6231
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	-0.0143	-0.0002	0.1356	0.6579	1.6459	0.0339	-0.0164	0.6434
VTP ($\eta = 2$)	-0.0153	0.0002	0.1300	0.6784	1.6618	0.0398	-0.0177	0.6607
VTP ($\eta = 3$)	-0.0165	0.0006	0.1268	0.6857	1.6561	0.0459	-0.0193	0.6647
VTP ($\eta = 4$)	-0.0182	0.0008	0.1253	0.6813	1.6434	0.0495	-0.0212	0.6584
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0290	0.0026**	0.2093	0.2406**	1.2121	0.0727	0.0246	0.2197**
VEIP ($\eta = 2$)	0.0434	0.0038**	0.2139	0.3027***	1.2661	0.0853	0.0382	0.2787***
VEIP ($\eta = 3$)	0.0649	0.0055**	0.2284	0.3778***	1.3341	0.0892	0.0595	0.3542***
VEIP ($\eta = 4$)	0.0979	0.0082**	0.2556	0.4666***	1.4274	0.0971	0.0920	0.4436***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	-0.0002	0.0000	0.2206	0.0959	1.0789	0.0395	-0.0026	0.0851
FTP	0.0000	0.0000	0.2198	0.0975	1.0804	0.0397	-0.0023	0.0867
EIP	0.0174	0.0015	0.2228	0.1744*	1.1413	0.0951	0.0117	0.1488
GMVP	0.0378	0.0035	0.2023	0.2926**	1.2558	0.0751	0.0333	0.2703**
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0410	0.0032	0.1186
EWI	-0.0186	-0.0015	0.2254	0.0125	1.0099	0.0000	-0.0186	0.0125
VTP ($\eta = 1$)	0.0151	0.0015	0.2058	0.1775	1.1479	0.0437	0.0125	0.1648
VTP ($\eta = 2$)	0.0242	0.0022	0.2046	0.2227	1.1869	0.0486	0.0213	0.2085
VTP ($\eta = 3$)	0.0324	0.0030*	0.2039	0.2638*	1.2228	0.0550	0.0291	0.2476*
VTP ($\eta = 4$)	0.0395	0.0036*	0.2036	0.2990*	1.2546	0.0629	0.0357	0.2804*

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

ITP, FTP, and GMVP, regardless of the dataset. Given that the VEIP consists of the GMVP and the VTB-mimicking portfolio, this result means that the VTB-mimicking portfolio does capture the superior returns from the volatility timing benchmark, as evidenced by the excess return and Jensen's alpha of the VEIP relative to the GMVP. Additionally, the only difference between the VEIP and the passive tracking portfolios (ITP and FTP) is volatility timing, this implies that the superior returns of the VEIP are mainly attributable to volatility timing. Furthermore, the VEIP outperforms the EIP in terms of excess returns and Jensen's alpha, especially when $\eta \in \{3, 4\}$. This is because the EIP is greatly impacted by parameter estimation errors of expected returns (Jiang *et al.* 2021) and lacks risk control (Wang *et al.* 2012). In addition, the VEIP also outperforms the EWP, EWI, and VTP in terms of excess return and Jensen's alpha, showing its advantages over the naïve diversification and simple active volatility timing strategies (Kirby and Ostdiek 2012).

Regarding standard deviations, when $\eta \in \{1, 2\}$, the VEIP is comparable to the benchmark index, the ITP, FTP, and EWP, outperforms the EIP and EWI, and underperforms the GMVP and VTP. When $\eta \in \{3, 4\}$, the VEIP has higher standard deviations than all benchmark portfolios except for the EIP, which is greatly affected by parameter estimation errors. This is because the volatility timing benchmark becomes more aggressive, as η rises, leading to the VEIP's high standard deviations. In terms of risk-adjusted returns, including the Sharpe ratio and omega ratio, the VEIP outperforms all benchmark portfolios, especially when $\eta \in \{3, 4\}$. For example, the VEIP's Sharpe ratio ranges from 0.7803 to 0.9217. The highest Sharpe ratio of the benchmark portfolio (EWI) based on the S&P 500 dataset is only 0.7691. This is because the VTB-mimicking portfolio component in the VEIP captures superior returns, and the GMVP component in the VEIP helps reduce risks, thereby boosting the VEIP's risk-adjusted returns.

Table 5. The out-of-sample performance of the examined portfolios with short sales.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0004	0.0003	0.1465	0.7092	1.6878	0.1261	− 0.0073	0.6585
VEIP ($\eta = 2$)	0.0227	0.0022*	0.1557	0.8108	1.7852	0.1790	0.0119	0.7415
VEIP ($\eta = 3$)	0.0549	0.0046**	0.1741	0.9099**	1.9050	0.1811	0.0440	0.8457*
VEIP ($\eta = 4$)	0.0659	0.0054**	0.1831	0.9248**	1.9251	0.1956	0.0540	0.8586**
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	− 0.0020	− 0.0002	0.1492	0.6802	1.6726	0.0340	− 0.0041	0.6671
FTP	− 0.0020	− 0.0002	0.1493	0.6798	1.6725	0.0342	− 0.0041	0.6666
EIP	0.0328	0.0024	0.1745	0.7811	1.7715	0.2825	0.0156	0.6849
GMVP	− 0.0149	0.0015	0.1278	0.6927	1.6475	0.1778	− 0.0257	0.6099
EWP	− 0.0114	− 0.0006	0.1449	0.6358	1.6311	0.0320	− 0.0133	0.6231
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	− 0.0143	− 0.0002	0.1356	0.6579	1.6459	0.0339	− 0.0164	0.6434
VTP ($\eta = 2$)	− 0.0153	0.0002	0.1300	0.6784	1.6618	0.0398	− 0.0177	0.6607
VTP ($\eta = 3$)	− 0.0165	0.0006	0.1268	0.6857	1.6561	0.0459	− 0.0193	0.6647
VTP ($\eta = 4$)	− 0.0182	0.0008	0.1253	0.6813	1.6434	0.0495	− 0.0212	0.6584
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0348	0.0031**	0.2059	0.2730**	1.2444	0.1207	0.0276	0.2377*
VEIP ($\eta = 2$)	0.0555	0.0048**	0.2170	0.3546***	1.3253	0.1593	0.0459	0.3104**
VEIP ($\eta = 3$)	0.0849	0.0072**	0.2382	0.4461***	1.4165	0.1816	0.0739	0.4002***
VEIP ($\eta = 4$)	0.1232	0.0103***	0.2698	0.5360***	1.5210	0.2019	0.1110	0.4908***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	− 0.0010	− 0.0001	0.2209	0.0925	1.0760	0.0400	− 0.0034	0.0816
FTP	− 0.0005	0.0000	0.2201	0.0951	1.0783	0.0400	− 0.0029	0.0842
EIP	0.0128	0.0010	0.2398	0.1426	1.1105	0.1859	0.0016	0.0961
GMVP	0.0567	0.0052*	0.2083	0.3748***	1.3479	0.1359	0.0485	0.3355***
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0410	0.0032	0.1186
EWI	− 0.0186	− 0.0015	0.2254	0.0125	1.0099	0.0000	− 0.0186	0.0125
VTP ($\eta = 1$)	0.0151	0.0015	0.2058	0.1775	1.1479	0.0437	0.0125	0.1648
VTP ($\eta = 2$)	0.0242	0.0022	0.2046	0.2227*	1.1869	0.0486	0.0213	0.2085
VTP ($\eta = 3$)	0.0324	0.0030*	0.2039	0.2638*	1.2228	0.0550	0.0291	0.2476*
VTP ($\eta = 4$)	0.0395	0.0036*	0.2036	0.2990**	1.2546	0.0629	0.0357	0.2804**

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Despite the VEIP's superior performance, a pivotal concern is whether its superior performance persists after accounting for transaction costs. To address this issue, we evaluate the transaction costs of the portfolios utilizing turnover, with the unit of transaction cost $c = 50\text{bps}$ (Kirby and Ostdiek 2012, Tu and Li 2024). Moreover, we consider their net excess return and net Sharpe ratio, which measure a strategy's performance after accounting for transaction costs.

The results in Table 4 show that the turnover of the VEIP without short sales ranges from 0.0727 to 0.1150 based on the two datasets, compared to 0 to 0.0880 for the benchmark portfolios with the exception of the EIP that has a turnover of 0.1377. This suggests that the VEIP incurs higher transaction costs than the benchmark portfolios, excluding the EIP. The primary reason is that the VEIP needs to adjust its exposure to the benchmark index in response to volatility timing factor changes, resulting in extensive weight adjustments and high turnover. However, the VEIP still outperforms all benchmark portfolios in terms of net excess returns and net Sharpe ratios

after accounting for transaction costs. For instance, based on the S&P 500 dataset, the VEIP's net excess returns range from 0.28% to 4.95%, while the highest net excess return among the benchmark portfolios is only 1.99% achieved by the EWI. Additionally, the VEIP achieves the highest net Sharpe ratio of 0.8852, surpassing the highest net Sharpe ratio of all the benchmark portfolios, which is 0.7691 achieved by the EWI. These findings confirm the VEIP's superior net excess returns and net risk-adjusted returns.

Collectively, the results in Table 4 confirm that the excess returns and risk-adjusted returns of the VEIP without short sales are higher than those of the benchmark portfolios, regardless of whether transaction costs are considered.

4.2.2. Out-of-sample performance with short sales. Table 5 reports the out-of-sample performance of the portfolios under consideration with short sales based on both datasets. Consistent with the results in Table 4, the results in Table 5 show that

Table 6. The market participation ratios of the examined portfolios.

	Without short sales			With short sales		
	UPR	DPR	PRD	UPR	DPR	PRD
Panel A: Participation ratios of the examined portfolios in the S&P 500 dataset						
Panel A.1: Participation ratios of the VEIP with $\rho = 14\%$						
VEIP ($\eta = 1$)	0.9827	0.9217	0.0610	0.9911	0.9820	0.0090
VEIP ($\eta = 2$)	1.0483	0.9174	0.1310	1.0538	0.9390	0.1147
VEIP ($\eta = 3$)	1.1361	0.9279	0.2081	1.1529	0.8918	0.2611
VEIP ($\eta = 4$)	1.1701	0.9160	0.2541	1.1808	0.8661	0.3146
Panel A.2: Participation ratios of the examined benchmark portfolios						
S&P 500	1.0000	1.0000	0.0000	1.0000	1.0000	0.0000
ITP	0.9949	1.0048	-0.0099	0.9949	1.0048	-0.0099
FTP	0.9952	1.0051	-0.0099	0.9952	1.0051	-0.0099
EIP	1.1461	1.0758	0.0703	1.1281	1.0491	0.0790
GMVP	0.7532	0.7168	0.0365	0.7665	0.7048	0.0617
EWP	0.9299	0.9572	-0.0273	0.9299	0.9572	-0.0273
EWI	1.0662	0.9788	0.0874	1.0662	0.9788	0.0874
VTP ($\eta = 1$)	0.8823	0.8963	-0.0140	0.8823	0.8963	-0.0140
VTP ($\eta = 2$)	0.8404	0.8322	0.0082	0.8404	0.8322	0.0082
VTP ($\eta = 3$)	0.8059	0.7819	0.0240	0.8059	0.7819	0.0240
VTP ($\eta = 4$)	0.7788	0.7471	0.0317	0.7788	0.7471	0.0317
Panel B: Participation ratios of the examined portfolios in the CSI 300 dataset						
Panel B.1: Participation ratios of the VEIP with $\rho = 18\%$						
VEIP ($\eta = 1$)	0.9597	0.8491	0.1106	0.9331	0.7987	0.1344
VEIP ($\eta = 2$)	1.0123	0.8525	0.1598	0.9952	0.7890	0.2063
VEIP ($\eta = 3$)	1.1092	0.8774	0.2318	1.1264	0.8219	0.3045
VEIP ($\eta = 4$)	1.2400	0.8963	0.3437	1.2884	0.8547	0.4337
Panel B.2: Participation ratios of the examined benchmark portfolios						
CSI 300	1.0000	1.0000	0.0000	1.0000	1.0000	0.0000
ITP	0.9837	0.9832	0.0005	0.9843	0.9865	-0.0022
FTP	0.9811	0.9793	0.0018	0.9815	0.9816	-0.0001
EIP	1.1678	0.9734	0.1944	1.2343	1.0896	0.1447
GMVP	0.8463	0.6940	0.1523	0.8181	0.5936	0.2245
EWP	0.9282	0.9013	0.0269	0.9282	0.9013	0.0269
EWI	0.9339	0.9974	-0.0635	0.9339	0.9974	-0.0635
VTP ($\eta = 1$)	0.9332	0.8717	0.0615	0.9332	0.8717	0.0615
VTP ($\eta = 2$)	0.9366	0.8419	0.0947	0.9366	0.8419	0.0947
VTP ($\eta = 3$)	0.9384	0.8134	0.1250	0.9384	0.8134	0.1250
VTP ($\eta = 4$)	0.9384	0.7871	0.1512	0.9384	0.7871	0.1512

Notes: This table reports the upside participation ratio (UPR), downside participation ratio (DPR), and participation ratio difference (PRD) of the examined portfolios relative to the benchmark index.

the VEIP with short sales outperforms the benchmark portfolios in terms of the excess return, Jensen's alpha, Sharpe ratio, omega ratio, net excess return, and net Sharpe ratio. For example, based on the CSI 300 dataset, the VEIP achieves the highest net Sharpe ratio of 0.4908, surpassing the highest net Sharpe ratio of all the benchmark portfolios, which is 0.3355 achieved by the GMVP. Moreover, the VEIP's standard deviation and turnover are higher than most benchmarks, except for the EIP. All these metrics rise with η , similar to the results in Table 4. Collectively, these results confirm the superior excess returns and risk-adjusted returns of the VEIP with short sales, regardless of whether transaction costs are considered.

Furthermore, we compare the out-of-sample performance of the VEIP with short sales with that without short sales. When short sales are allowed, the VEIP has higher excess returns and Jensen's alpha, particularly for $\eta \in \{2, 3, 4\}$. This indicates that short sales enhance the VEIP's returns. The reasons are twofold. First, the VEIP can earn extra returns

through leverage when short sales are allowed (Clarke *et al.* 2008); Second, the VEIP avoids the impacts of parameter estimation errors of expected returns. However, as Table 5 shows, short sales also increase the standard deviation and turnover of the VEIP. This aligns with Kirby and Ostdiek's (2012) findings that short sales can lead to extreme weights and amplify the impact of parameter estimation errors, thereby leading to higher standard deviations and turnovers. Similar patterns are observed in the EIP and GMVP. In terms of the Sharpe ratio, omega ratio, and net Sharpe ratio, the VEIP with short sales shows mixed results based on the two datasets. Based on the CSI 300 dataset, it outperforms its counterpart without short sales, but this is not true based on the S&P 500 dataset. This finding corroborates the finding of Clarke *et al.* (2008), which suggests that a 130/30 strategy better captures returns with leverage in highly volatile markets.

Additionally, we compare the VEIP's out-of-sample performance in Tables 4 and 5 with the in-sample performance of the volatility timing benchmark in Table 3. The VEIP's

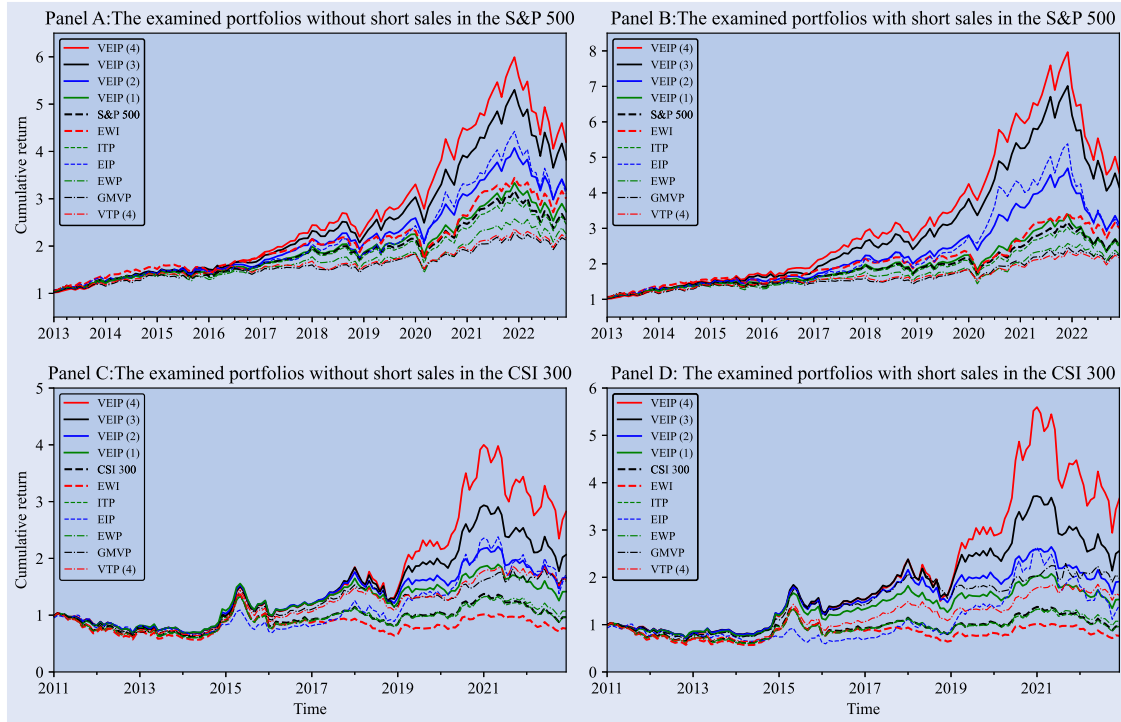


Figure 2. The cumulative return curves for the examined portfolios. This figure shows the out-of-sample cumulative return curves for different portfolios without and with short sales in the S&P 500 (2013:01–2022:12) and the CSI 300 (2011:01–2022:12) datasets, where ρ is 14% for the S&P 500 dataset and 18% for the CSI 300 dataset. We have excluded the cumulative return curves of the FTP and the VTP ($\eta = 1, 2, 3$), as the former closely mirrors the benchmark index and the latter underperforms the VTP ($\eta = 4$). The number # in parentheses of the VEIP (#) and VTP (#) is η .

excess returns and Sharpe ratios are lower than the volatility timing benchmark. This means the VEIP does not fully capture the benchmark's returns, regardless of whether short sales are allowed. The primary reasons are twofold. First, the volatility timing benchmark is an in-sample portfolio that avoids estimation errors and uses high leverage. The VEIP, however, is affected by estimation errors and cannot perfectly match this leverage. Second, the correlations between the volatility timing benchmark's returns and those of risky assets can affect the tracking effect (Jiang *et al.* 2021). When the benchmark has a high correlation with a risky asset, the VEIP captures its returns better. Conversely, it captures less. Despite this, the VEIP still outperforms the benchmark portfolios, as evidenced in Tables 4 and 5.

Overall, the findings in Tables 3–5 confirm the VEIP's superior excess returns and risk-adjusted returns before and after considering transaction costs, demonstrating the economic value of volatility timing in enhanced indexation, regardless of whether short sales are allowed.

4.3. Upside participation and downside participation

To better understand why the VEIP can generate superior performance, we explore its ability to capture upside returns and mitigate downside losses, using the upside participation ratio (UPR) and downside participation ratio (DPR) (Qian 2015).

The UPR of a portfolio, using the benchmark index as a baseline, is defined as:

$$UPR = \frac{E(r_P | r_I > 0)}{E(r_I | r_I > 0)}, \quad (13)$$

where r_I is the benchmark index return, and r_P is the out-of-sample return of the examined portfolio.

$UPR > 1$ ($UPR < 1$) implies that the portfolio captures higher (lower) upside returns than the benchmark index when the benchmark index rises. Similarly, the DPR of the portfolio is computed as:

$$DPR = \frac{E(r_P | r_I < 0)}{E(r_I | r_I < 0)} \quad (14)$$

$DPR < 1$ ($DPR > 1$) indicates that the portfolio mitigates higher (lower) losses than the benchmark index when the benchmark index declines. Considering both the UPR and DPR, Qian (2015) computes the participation ratio difference (PRD) as follows:

$$PRD = UPR - DPR \quad (15)$$

$PRD > 0$ ($PRD < 0$) implies that the portfolio outperforms (underperforms) the benchmark index in terms of overall market participation.

Table 6 reports the UPR, DPR, and PRD of the portfolios considered without and with short sales based on both datasets. We observe that the UPR, DPR, and PRD of the ITP and FTP are all close to those of the benchmark index. This is because both the ITP and FTP aim to replicate the benchmark index's performance, thereby achieving similar participation ratios. Furthermore, the VEIP outperforms the benchmark index, ITP, and FTP in terms of the UPR, DPR, and PRD, regardless of whether short sales are allowed, particularly when $\eta \in \{2, 3, 4\}$. This suggests that, on average, the

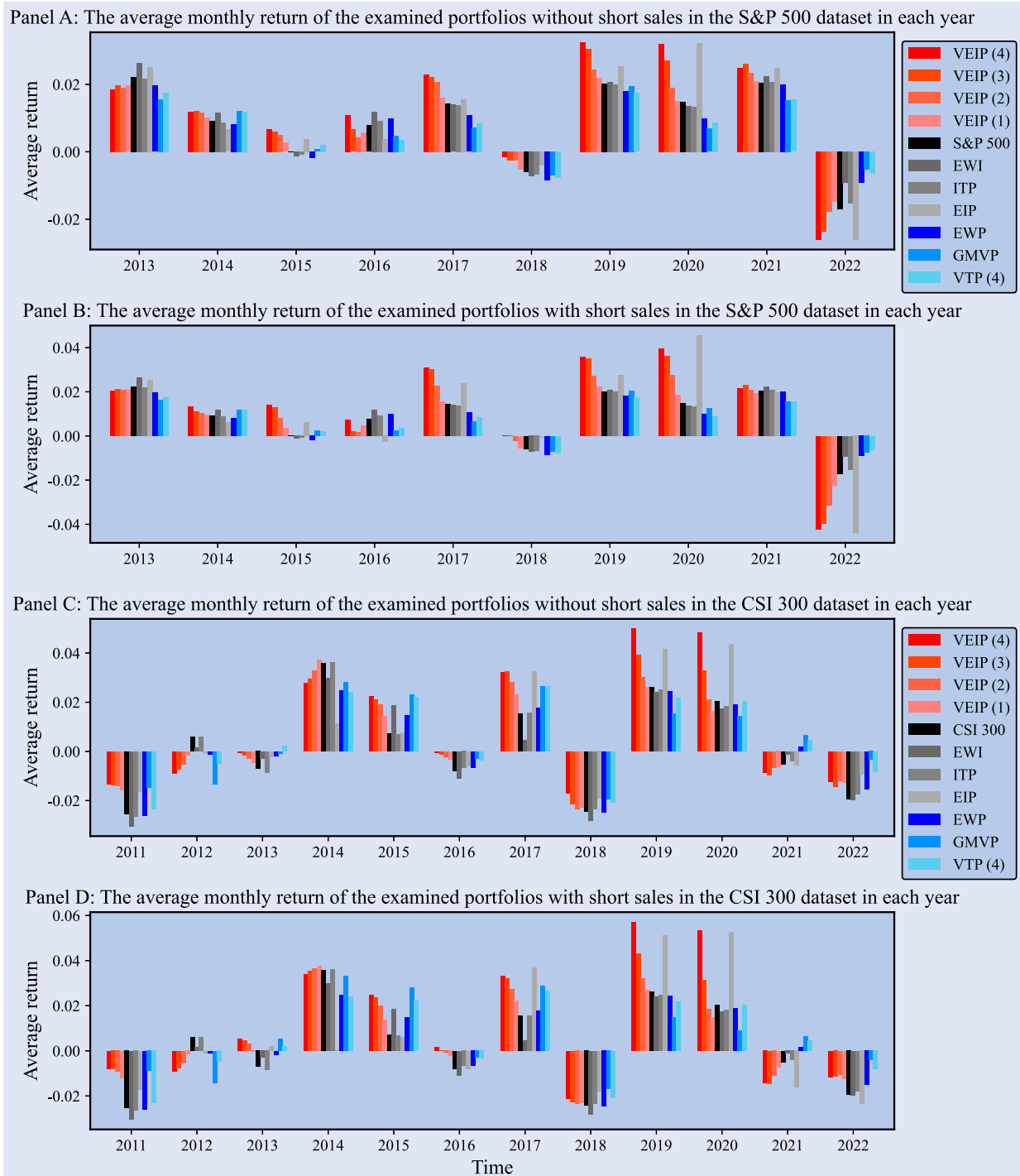


Figure 3. The average monthly return of the examined portfolios in each year. This figure shows the average monthly return per year for different portfolios without and with short sales for the S&P 500 (2013:01–2022:12) and the CSI 300 (2011:01–2022:12) datasets, where ρ is 14% for the S&P 500 dataset and 18% for the CSI 300 dataset. For clarity, we have excluded the results of the FTP and the VTP ($\eta = 1, 2, 3$), as the former closely mirrors the benchmark index while the latter underperforms the VTP ($\eta = 4$). The number # in parentheses of the VEIP (#) and VTP (#) is η .

VEIP captures higher returns than the benchmark index when it rises and mitigates losses in adverse situations. This reveals why the VEIP can obtain superior returns than the ITP, FTP, and benchmark index, and confirms that investors can obtain superior returns with volatility timing.

Next, compared to the EIP, the VEIP achieves comparable UPR and better DPR based on both datasets, regardless of whether short sales are allowed. This is because the EIP

while striving to maximize excess returns over the benchmark index, lacks risk-controlling approaches (Huang *et al.* 2018). In contrast, the VEIP can capture returns and mitigate losses via volatility timing. Therefore, the VEIP achieves a higher PRD than the EIP due to its better DPR. Subsequently, compared to the GMVP and VTP, the VEIP yields higher UPR but lower DPR based on both datasets. This aligns with the results of Jiang *et al.* (2021) that the GMVP has good DPR

Table 7. Out-of-sample performance of the volatility timing benchmark.

	ER	STD	Sharpe	Omega	Alpha _{noshort}	Alpha _{short}
Panel A: Out-of-sample performance of the volatility timing benchmark for the S&P 500 index ($\rho = 14\%$)						
VTB ($\eta = 1$)	-0.0230	0.1357	0.6659	1.6273	0.0020	0.0012
VTB ($\eta = 2$)	-0.0356	0.1367	0.6145	1.5957	0.0044*	0.0043*
VTB ($\eta = 3$)	-0.0410	0.1373	0.5645	1.5620	0.0069**	0.0078**
VTB ($\eta = 4$)	-0.0430	0.1380	0.5182	1.5275	0.0086**	0.0094***
Panel B: Out-of-sample performance of the volatility timing benchmark for the CSI 300 index ($\rho = 18\%$)						
VTB ($\eta = 1$)	0.0075	0.1969	0.1842	1.1613	0.0012	0.0018
VTB ($\eta = 2$)	0.0079	0.1842	0.2069	1.1963	0.0023	0.0033
VTB ($\eta = 3$)	0.0070	0.1764	0.2129	1.2168	0.0042*	0.0058*
VTB ($\eta = 4$)	0.0058	0.1717	0.2166	1.2357	0.0070**	0.0090**

Notes: VTB is the volatility timing benchmark. ER, STD, Sharpe, and Omega represent the annualized excess return, annualized standard deviation, Sharpe ratio, and omega ratio, respectively. Alpha_{noshort} is the spanning regression alpha of the out-of-sample returns of the VEIP without short sales on the out-of-sample returns of the volatility timing benchmark. Alpha_{short} is the spanning regression alpha of the out-of-sample returns of the VEIP with short sales on the out-of-sample returns of the volatility timing benchmark. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

but lower UPR. This is because the low beta of the GMVP and VTP helps them reduce portfolio risks, but limits their upside returns (Bednarek and Patel 2018). Therefore, benefiting from the UPR, the VEIP obtains higher PRDs than the GMVP and VTP. Moreover, as the VEIP consists of the GMVP and the VTB-mimicking portfolio, this result indicates that the VTB-mimicking portfolio earns higher returns than the GMVP when the benchmark rises. This finding is consistent with that in Tables 4 and 5.

Finally, compared to the EWP and EWI, the VEIP obtains better UPR, DPR, and PRD, particularly when $\eta \in \{3, 4\}$. Additionally, as η rises, the PRD of the VEIP tends to perform better, explaining why all the performance measures of the VEIP rise with η . Overall, these results demonstrate the superior upside participation and downside participation of the VEIP achieved through volatility timing, and explain why the VEIP generates superior excess returns and risk-adjusted returns.

4.4. Portfolio cumulative returns over time

To further explore the performance of the VEIP over time, Figure 2 displays the out-of-sample cumulative returns of all the portfolios considered without and with short sales based on both datasets. The VEIP outperforms all benchmarks in terms of cumulative returns over time. Moreover, its cumulative returns increase with η , matching the results in Tables 4 and 5.

However, Figure 2 reveals the VEIP's undesirable drawdown, particularly for $\eta \in \{3, 4\}$, underscoring its potential limitations. To further investigate, in Figure 3, we plot the average monthly returns per year for the portfolios under consideration. We note that in certain years, such as 2022 for the S&P 500 as well as 2012 and 2021 for the CSI 300, the VEIP underperforms the benchmark index in average returns. These are due to two reasons. First, volatility treats upside returns and downside losses symmetrically (Huang *et al.* 2022a). In this context, a volatility timing strategy can increase exposure to the benchmark index when volatility is low and the

benchmark index is down, thereby leading to potential losses. On the other hand, it might also reduce exposure to the benchmark index when volatility is high and the benchmark index is up, thereby reducing potential returns (Albeverio *et al.* 2013). For example, in 2012, China's stock market had a low-volatility downturn for the first 11 months, as shown in Figure 1, resulting in the VEIP's losses. Similarly, the EIP, GMVP, and VTP also suffer from losses in that year.

Second, as pointed out by Yang and Huang (2022), unexpected extreme events can cause significant fluctuations in the stock market, rendering historical data ineffective in predicting future returns and risks. Moreover, Wang and Yan (2021) suggest that the efficacy of volatility timing mainly stems from the persistence of volatility. However, abrupt changes in returns can undermine the persistence of volatility, reducing the performance of volatility timing strategies. For example, major events such as China's real estate policy adjustments in 2021 and the Federal Reserve's interest rate hikes along with the Russia-Ukraine conflict in 2022 caused sudden fluctuations in the Chinese and US stock markets, respectively. Consequently, when focusing on the 2021 CSI 300 and 2022 S&P 500 datasets, the VEIP underperforms the benchmark index in terms of average returns because of the impact of these extreme events. Moreover, Figure 3 shows that using a smaller η enables the VEIP to obtain comparable returns to the benchmark index during these years, mitigating its potential limitations.

Despite these limitations, the VEIP maintains its overall superior performance. Figure 3 shows that in most years, especially from 2014 to 2021, based on the S&P 500 dataset, the VEIP outperforms the benchmark index. More importantly, it incurs lower losses during negative index returns and achieves higher returns during positive index returns, consistent with the results in Table 6. These findings reaffirm the VEIP's superior out-of-sample returns.

Overall, while Figures 2 and 3 reveal the VEIP's limitations, the results in Tables 4 and 5, as well as Figures 2 and 3, demonstrate the overall benefits of the VEIP, regardless of whether short sales are allowed.

Table 8. Out-of-sample performance of the VEIP with different ρ in the S&P 500 dataset.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the VEIP without short sales								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 12\%$								
VEIP ($\eta = 1$)	0.0024	0.0010	0.1379	0.7678	1.7610	0.0729	-0.0021	0.7374
VEIP ($\eta = 2$)	0.0141	0.0019*	0.1401	0.8397	1.8341	0.0772	0.0094	0.8070
VEIP ($\eta = 3$)	0.0282	0.0025**	0.1524	0.8639*	1.8809	0.1089	0.0216	0.8201*
VEIP ($\eta = 4$)	0.0480	0.0036**	0.1683	0.9000**	1.9527	0.1105	0.0413	0.8592**
Panel A.2: Out-of-sample performance of the VEIP with $\rho = 16\%$								
VEIP ($\eta = 1$)	0.0121	0.0012*	0.1470	0.7864	1.7963	0.0823	0.0070	0.7535
VEIP ($\eta = 2$)	0.0324	0.0022**	0.1621	0.8384	1.8674	0.0958	0.0266	0.8028
VEIP ($\eta = 3$)	0.0538	0.0038**	0.1717	0.9165**	1.9574	0.1127	0.0470	0.8767**
VEIP ($\eta = 4$)	0.0598	0.0041**	0.1781	0.9171***	1.9496	0.0986	0.0539	0.8830***
Panel B: Out-of-sample performance of the VEIP with short sales								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 12\%$								
VEIP ($\eta = 1$)	-0.0016	0.0005	0.1418	0.7185	1.6945	0.1231	-0.0091	0.6675
VEIP ($\eta = 2$)	0.0100	0.0017	0.1450	0.7826	1.7371	0.1841	-0.0012	0.7063
VEIP ($\eta = 3$)	0.0352	0.0037*	0.1592	0.8712*	1.8376	0.1723	0.0248	0.8049*
VEIP ($\eta = 4$)	0.0539	0.0047*	0.1762	0.8932***	1.8849	0.1934	0.0422	0.8255*
Panel B.2: Out-of-sample performance of the VEIP with $\rho = 16\%$								
VEIP ($\eta = 1$)	0.0021	0.0002	0.1518	0.6960	1.6710	0.1323	-0.0059	0.6444
VEIP ($\eta = 2$)	0.0352	0.0024	0.1698	0.8170	1.8080	0.1880	0.0238	0.7502
VEIP ($\eta = 3$)	0.0633	0.0050*	0.1827	0.9130***	1.9040	0.1902	0.0517	0.8496*
VEIP ($\eta = 4$)	0.0674	0.0052*	0.1891	0.9040***	1.9030	0.1769	0.0567	0.8467**

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 9. Out-of-sample performance of the VEIP with different ρ in the CSI 300 dataset.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the VEIP without short sales								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 16\%$								
VEIP ($\eta = 1$)	0.0310	0.0028**	0.2074	0.2528*	1.2229	0.0735	0.0266	0.2315*
VEIP ($\eta = 2$)	0.0427	0.0038**	0.2099	0.3055***	1.2677	0.0838	0.0377	0.2815**
VEIP ($\eta = 3$)	0.0595	0.0051**	0.2206	0.3668***	1.3255	0.0906	0.0541	0.3421***
VEIP ($\eta = 4$)	0.0869	0.0073**	0.2451	0.4417***	1.4000	0.0909	0.0813	0.4194***
Panel A.2: Out-of-sample performance of the VEIP with $\rho = 20\%$								
VEIP ($\eta = 1$)	0.0238	0.0021*	0.2106	0.2148*	1.1876	0.0752	0.0193	0.1933*
VEIP ($\eta = 2$)	0.0422	0.0036**	0.2176	0.2921**	1.2559	0.0918	0.0366	0.2667**
VEIP ($\eta = 3$)	0.0626	0.0053**	0.2313	0.3632***	1.3208	0.1053	0.0563	0.3358***
VEIP ($\eta = 4$)	0.0945	0.0079**	0.2632	0.4405***	1.4002	0.1016	0.0885	0.4170***
Panel B: Out-of-sample performance of the VEIP with short sales								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 16\%$								
VEIP ($\eta = 1$)	0.0375	0.0034*	0.2039	0.2886*	1.2600	0.1251	0.0299	0.2517*
VEIP ($\eta = 2$)	0.0562	0.0049**	0.2126	0.3653***	1.3359	0.1524	0.0471	0.3221***
VEIP ($\eta = 3$)	0.0756	0.0065**	0.2255	0.4300***	1.4013	0.1691	0.0653	0.3848***
VEIP ($\eta = 4$)	0.1071	0.0091**	0.2560	0.5019***	1.4796	0.1894	0.0956	0.4575***
Panel B.2: Out-of-sample performance of the VEIP with $\rho = 20\%$								
VEIP ($\eta = 1$)	0.0318	0.0028**	0.2103	0.2530*	1.2244	0.1154	0.0249	0.2200*
VEIP ($\eta = 2$)	0.0546	0.0047**	0.2242	0.3388***	1.3089	0.1684	0.0444	0.2936**
VEIP ($\eta = 3$)	0.0867	0.0073***	0.2452	0.4410***	1.4132	0.1992	0.0747	0.3920***
VEIP ($\eta = 4$)	0.1159	0.0096***	0.2796	0.4912***	1.4659	0.2146	0.1030	0.4447***

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 10. Out-of-sample performance of the examined portfolios with the shrinkage estimator without short sales.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0096	0.0012*	0.1432	0.7898	1.7985	0.0735	0.0051	0.7599
VEIP ($\eta = 2$)	0.0252	0.0021**	0.1519	0.8469*	1.8623	0.0935	0.0195	0.8101
VEIP ($\eta = 3$)	0.0460	0.0033**	0.1668	0.8962**	1.9437	0.1137	0.0391	0.8543**
VEIP ($\eta = 4$)	0.0565	0.0040**	0.1729	0.9250***	1.9852	0.1013	0.0503	0.8889**
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	-0.0008	-0.0002	0.1501	0.6842	1.6772	0.0328	-0.0028	0.6717
FTP	-0.0026	-0.0002	0.1481	0.6815	1.6749	0.0321	-0.0046	0.6690
EIP	0.0254	0.0011	0.1764	0.7309	1.7126	0.1332	0.0109	0.6835
GMVP	-0.0199	0.0010	0.1237	0.6758	1.6362	0.0766	-0.0246	0.6392
EWP	-0.0114	-0.0006	0.1449	0.6358	1.6311	0.0320	-0.0133	0.6231
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	-0.0143	-0.0002	0.1356	0.6579	1.6459	0.0339	-0.0164	0.6434
VTP ($\eta = 2$)	-0.0153	0.0002	0.1300	0.6784	1.6618	0.0398	-0.0177	0.6607
VTP ($\eta = 3$)	-0.0165	0.0006	0.1268	0.6857	1.6561	0.0459	-0.0193	0.6647
VTP ($\eta = 4$)	-0.0182	0.0008	0.1253	0.6813	1.6434	0.0495	-0.0212	0.6584
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0276	0.0025**	0.2093	0.2342*	1.2059	0.0725	0.0233	0.2133*
VEIP ($\eta = 2$)	0.0436	0.0038**	0.2143	0.3032***	1.2661	0.0837	0.0385	0.2798**
VEIP ($\eta = 3$)	0.0652	0.0056**	0.2290	0.3781***	1.3338	0.0886	0.0599	0.3548***
VEIP ($\eta = 4$)	0.0979	0.0082**	0.2561	0.4658***	1.4258	0.0964	0.0921	0.4430***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	-0.0031	-0.0002	0.2226	0.0822	1.0676	0.0387	-0.0054	0.0718
FTP	0.0009	0.0001	0.2168	0.1027	1.0850	0.0403	-0.0016	0.0916
EIP	0.0152	0.0013	0.2233	0.1640	1.1323	0.0936	0.0096	0.1388
GMVP	0.0404	0.0038	0.2023	0.3055***	1.2678	0.0739	0.0360	0.2836**
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0410	0.0032	0.1186
EWI	-0.0186	-0.0015	0.2254	0.0125	1.0099	0.0000	-0.0186	0.0125
VTP ($\eta = 1$)	0.0151	0.0015	0.2058	0.1775	1.1479	0.0437	0.0125	0.1648
VTP ($\eta = 2$)	0.0242	0.0022	0.2046	0.2227*	1.1869	0.0486	0.0213	0.2085
VTP ($\eta = 3$)	0.0324	0.0030*	0.2039	0.2638*	1.2228	0.0550	0.0291	0.2476*
VTP ($\eta = 4$)	0.0395	0.0036**	0.2036	0.2990**	1.2546	0.0629	0.0357	0.2804**

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

4.5. Additional analysis

Prior results confirm the VEIP's superior performance compared to the benchmark portfolios considered. However, it is not clear whether the VEIP adds to the overall performance relative to the volatility timing benchmark in practice. To investigate this, following Barroso and Santa-Clara (2015), we compute the out-of-sample performance of the volatility timing benchmark, using the rolling window approach described in Section 3, with a six-month parameter estimation window and a one-month rebalancing frequency. In addition, following Albeverio *et al.* (2013), we set the volatility timing factor to be no more than 1.5, owing to the impractically high leverage of the volatility timing benchmark.[†]

[†] We also set the volatility timing factor to be no more than 1.3 and 2.0, and the results still hold. Notably, setting a leverage cap aligns with regulatory requirements for mutual funds, such as the UCITS-compliant funds.

Table 7 reports the out-of-sample performance of the volatility timing benchmark. When comparing these results with the results of the VEIP reported in Tables 4 and 5, we find that the VEIP outperforms the volatility timing benchmark in terms of the excess return, Sharpe ratio, and omega ratio based on both datasets, regardless of whether short sales are allowed. It is noteworthy that the VEIP also demonstrates a significant alpha relative to the volatility timing benchmark. For example, based on the S&P 500 dataset, the $\text{Alpha}_{\text{short}}$ achieves 0.0094, which is significant at the 1% level, when short sales are allowed. This is due to two reasons. First, the volatility timing benchmark used in the VEIP is an in-sample portfolio, which mitigates the impact of parameter estimation errors. However, when investors implement the volatility timing benchmark in practice, the out-of-sample performance of the volatility timing benchmark is inevitably greatly affected by parameter estimation errors. Second, by tracking the in-sample volatility timing benchmark, the VEIP can capture

Table 11. Out-of-sample performance of the examined portfolios with the shrinkage estimator with short sales.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0022	0.0005	0.1473	0.7173	1.6978	0.1156	-0.0049	0.6712
VEIP ($\eta = 2$)	0.0262	0.0024	0.1575	0.8233	1.8018	0.1774	0.0154	0.7554
VEIP ($\eta = 3$)	0.0569	0.0047**	0.1751	0.9159**	1.9136	0.1760	0.0462	0.8542*
VEIP ($\eta = 4$)	0.0657	0.0053**	0.1832	0.9233***	1.9255	0.1957	0.0538	0.8575**
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	-0.0008	-0.0002	0.1501	0.6842	1.6772	0.0328	-0.0028	0.6717
FTP	-0.0026	-0.0002	0.1481	0.6815	1.6749	0.0321	-0.0046	0.6690
EIP	0.0323	0.0023	0.1770	0.7675	1.7521	0.2607	0.0164	0.6799
GMVP	-0.0162	0.0015	0.1270	0.6875	1.6400	0.1531	-0.0255	0.6160
EWP	-0.0114	-0.0006	0.1449	0.6358	1.6311	0.0320	-0.0133	0.6231
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	-0.0143	-0.0002	0.1356	0.6579	1.6459	0.0339	-0.0164	0.6434
VTP ($\eta = 2$)	-0.0153	0.0002	0.1300	0.6784	1.6618	0.0398	-0.0177	0.6607
VTP ($\eta = 3$)	-0.0165	0.0006	0.1268	0.6857	1.6561	0.0459	-0.0193	0.6647
VTP ($\eta = 4$)	-0.0182	0.0008	0.1253	0.6813	1.6434	0.0495	-0.0212	0.6584
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0362	0.0032**	0.2069	0.2785*	1.2496	0.1093	0.0296	0.2467*
VEIP ($\eta = 2$)	0.0569	0.0049**	0.2168	0.3612***	1.3314	0.1489	0.0479	0.3199***
VEIP ($\eta = 3$)	0.0855	0.0072***	0.2389	0.4474***	1.4172	0.1738	0.0750	0.4036***
VEIP ($\eta = 4$)	0.1223	0.0102***	0.2718	0.5288***	1.5126	0.1891	0.1109	0.4868***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	-0.0034	-0.0003	0.2228	0.0807	1.0663	0.0396	-0.0058	0.0700
FTP	0.0009	0.0001	0.2168	0.1027	1.0850	0.0403	-0.0016	0.0916
EIP	0.0117	0.0009	0.2407	0.1374	1.1063	0.1859	0.0005	0.0911
GMVP	0.0598	0.0054*	0.2083	0.3896***	1.3616	0.1324	0.0518	0.3513***
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0410	0.0032	0.1186
EWI	-0.0186	-0.0015	0.2254	0.0125	1.0099	0.0000	-0.0186	0.0125
VTP ($\eta = 1$)	0.0151	0.0015	0.2058	0.1775	1.1479	0.0437	0.0125	0.1648
VTP ($\eta = 2$)	0.0242	0.0022	0.2046	0.2227	1.1869	0.0486	0.0213	0.2085
VTP ($\eta = 3$)	0.0324	0.0030*	0.2039	0.2638*	1.2228	0.0550	0.0291	0.2476**
VTP ($\eta = 4$)	0.0395	0.0036*	0.2036	0.2990*	1.2546	0.0629	0.0357	0.2804**

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

the superior returns achieved through high leverage. However, it is hard for investors to implement a volatility timing benchmark with so high leverage in practice, especially when $\eta \in \{3, 4\}$.

Overall, these findings validate the VEIP's superior out-of-sample performance compared to that of the volatility timing benchmark, emphasizing the added value of the proposed strategy relative to the volatility timing benchmark in practice.

4.6. Robustness tests

In this section, we perform robustness tests of the empirical results of the VEIP under different target volatilities, different parameter estimation methods, and different parameter estimation windows.

First, we evaluate the VEIP's out-of-sample performance with alternative target volatilities. Specifically, we consider ρ values of 12% and 16% for the S&P 500 and 16% and 20% for the CSI 300. Tables 8 and 9 reveal that the VEIP consistently outperforms all benchmark portfolios (as referenced in Tables 4 and 5) in terms of excess returns and risk-adjusted returns across these various ρ values.

Second, we assess the VEIP's out-of-sample performance using Ledoit and Wolf's (2003) shrinkage covariance matrix estimator. Tables 10 and 11 demonstrate that the VEIP, when utilizing the shrinkage estimator, consistently outperforms all benchmark portfolios in terms of excess returns and risk-adjusted returns.

Third, we reassess the VEIP's out-of-sample performance with a narrower estimation window of 72 months. Tables 12 and 13 indicate that the VEIP, using the 72-month estimation

Table 12. Out-of-sample performance of the examined portfolios without short sales using a 72-month estimation window.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0147	0.0015*	0.1447	0.8174	1.8238	0.0975	0.0088	0.7778
VEIP ($\eta = 2$)	0.0325	0.0025**	0.1551	0.8766*	1.8908	0.1264	0.0248	0.8272*
VEIP ($\eta = 3$)	0.0462	0.0032**	0.1672	0.8950**	1.9314	0.1362	0.0380	0.8439**
VEIP ($\eta = 4$)	0.0582	0.0041**	0.1721	0.9398***	1.9803	0.1627	0.0484	0.8820***
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	-0.0009	-0.0001	0.1491	0.6883	1.6807	0.0349	-0.0030	0.6748
FTP	-0.0008	-0.0001	0.1495	0.6874	1.6803	0.0349	-0.0029	0.6739
EIP	0.0380	0.0022	0.1769	0.7995	1.7940	0.1357	0.0201	0.7527
GMVP	-0.0196	0.0009	0.1252	0.6703	1.6396	0.1192	-0.0268	0.6141
EWP	-0.0114	-0.0006	0.1449	0.6358	1.6311	0.0237	-0.0133	0.6261
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	-0.0140	-0.0002	0.1358	0.6592	1.6454	0.0364	-0.0162	0.6437
VTP ($\eta = 2$)	-0.0148	0.0002	0.1304	0.6799	1.6617	0.0480	-0.0178	0.6585
VTP ($\eta = 3$)	-0.0161	0.0005	0.1273	0.6865	1.6580	0.0589	-0.0197	0.6595
VTP ($\eta = 4$)	-0.0179	0.0007	0.1258	0.6810	1.6434	0.0669	-0.0219	0.6499
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0278	0.0025*	0.2076	0.2369*	1.2074	0.1018	0.0216	0.2075*
VEIP ($\eta = 2$)	0.0438	0.0038**	0.2137	0.3049**	1.2667	0.1317	0.0358	0.2679**
VEIP ($\eta = 3$)	0.0795	0.0068***	0.2294	0.4399***	1.4048	0.1449	0.0707	0.4018***
VEIP ($\eta = 4$)	0.1038	0.0088***	0.2453	0.5102***	1.4779	0.1468	0.0949	0.4740***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	0.0005	0.0001	0.2213	0.0990	1.0817	0.0410	-0.0019	0.0879
FTP	0.0010	0.0001	0.2204	0.1014	1.0837	0.0415	-0.0015	0.0901
EIP	0.0464	0.0040	0.2526	0.2683*	1.2236	0.2363	0.0016	0.2121
GMVP	0.0268	0.0026	0.2030	0.2372*	1.2034	0.1051	0.0204	0.2061
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0340	0.0032	0.1206
EWI	-0.0186	-0.0015	0.2254	0.0125	1.0099	0.0000	-0.0186	0.0125
VTP ($\eta = 1$)	0.0181	0.0017	0.2048	0.1929	1.1610	0.0486	0.0152	0.1787
VTP ($\eta = 2$)	0.0294	0.0027	0.2037	0.2492*	1.2105	0.0611	0.0257	0.2312
VTP ($\eta = 3$)	0.0387	0.0035*	0.2044	0.2942*	1.2518	0.0762	0.0341	0.2717*
VTP ($\eta = 4$)	0.0459	0.0041**	0.2063	0.3260**	1.2821	0.0916	0.0403	0.2993**

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

window, outperforms all benchmark portfolios in terms of excess returns and risk-adjusted returns.

Additionally, the VEIP exhibits superior participation ratios and cumulative returns across different robustness tests. These results are not detailed for brevity.

5. Conclusions

This paper develops a novel volatility timing-based enhanced indexation strategy, which minimizes the variance of the differences in returns between a tracking portfolio and a volatility timing benchmark. We establish the volatility timing-based enhanced indexation model and derive the optimal portfolio. The VEIP consists of the GMVP and a VTb-mimicking portfolio that captures the volatility timing benchmark returns. Using the US and Chinese stock market

data, we confirm that the VEIP outperforms the benchmark index and seven other benchmark portfolios in terms of excess returns, risk-adjusted returns, and cumulative returns, irrespective of short sales. Despite higher transaction costs, the VEIP attains superior net Sharpe ratios. The superior performance is due to the VEIP's better upside participation and downside participation obtained from volatility timing. These findings remain robust across various target volatilities, parameter estimation methods, and parameter estimation windows.

However, the VEIP is not free of drawbacks. In the case of a falling market accompanied by low volatility, volatility timing might increase the VEIP's market exposure, resulting in potential losses. In addition, unexpected extreme events may also affect the out-of-sample performance of the VEIP. These drawbacks can be mitigated by selecting smaller volatility timing parameters and do not eliminate the overall economic

Table 13. Out-of-sample performance of the examined portfolios with short sales using a 72-month estimation window.

	ER	Alpha	STD	SR	Omega	Turnover	NER	NSR
Panel A: Out-of-sample performance of the examined portfolios in the S&P 500 dataset								
Panel A.1: Out-of-sample performance of the VEIP with $\rho = 14\%$								
VEIP ($\eta = 1$)	0.0137	0.0014*	0.1481	0.7916	1.7873	0.1597	0.0040	0.7280
VEIP ($\eta = 2$)	0.0427	0.0036**	0.1601	0.9131**	1.9223	0.2369	0.0283	0.8238
VEIP ($\eta = 3$)	0.0609	0.0050**	0.1741	0.9444**	1.9523	0.2201	0.0476	0.8667*
VEIP ($\eta = 4$)	0.0721	0.0059**	0.1782	0.9849***	2.0090	0.2576	0.0564	0.8972**
Panel A.2: Out-of-sample performance of the examined benchmark portfolios								
S&P 500	0.0000	0.0000	0.1479	0.6995	1.6899	0.0000	0.0000	0.6995
ITP	−0.0009	−0.0001	0.1491	0.6883	1.6807	0.0349	−0.0030	0.6748
FTP	−0.0008	−0.0001	0.1495	0.6874	1.6803	0.0349	−0.0029	0.6739
EIP	0.0498	0.0037	0.1893	0.8100	1.7878	0.2687	0.0333	0.7230
GMVP	−0.0103	0.0018	0.1276	0.7306	1.7022	0.2289	−0.0242	0.6237
EWP	−0.0114	−0.0006	0.1449	0.6358	1.6311	0.0237	−0.0133	0.6261
EWI	0.0199	0.0012	0.1605	0.7691	1.8254	0.0000	0.0199	0.7691
VTP ($\eta = 1$)	−0.0140	−0.0002	0.1358	0.6592	1.6454	0.0364	−0.0162	0.6437
VTP ($\eta = 2$)	−0.0148	0.0002	0.1304	0.6799	1.6617	0.0480	−0.0178	0.6585
VTP ($\eta = 3$)	−0.0161	0.0005	0.1273	0.6865	1.6580	0.0589	−0.0197	0.6595
VTP ($\eta = 4$)	−0.0179	0.0007	0.1258	0.6810	1.6434	0.0669	−0.0219	0.6499
Panel B: Out-of-sample performance of the examined portfolios in the CSI 300 dataset								
Panel B.1: Out-of-sample performance of the VEIP with $\rho = 18\%$								
VEIP ($\eta = 1$)	0.0339	0.0030*	0.2110	0.2624*	1.2328	0.1689	0.0237	0.2142*
VEIP ($\eta = 2$)	0.0567	0.0049**	0.2236	0.3492**	1.3175	0.2239	0.0431	0.2891**
VEIP ($\eta = 3$)	0.0987	0.0083**	0.2456	0.4891**	1.4700	0.2473	0.0837	0.4286**
VEIP ($\eta = 4$)	0.1329	0.0111***	0.2637	0.5850***	1.5870	0.2657	0.1167	0.5243***
Panel B.2: Out-of-sample performance of the examined benchmark portfolios								
CSI 300	0.0000	0.0000	0.2221	0.0962	1.0791	0.0000	0.0000	0.0962
ITP	0.0000	0.0000	0.2215	0.0964	1.0794	0.0424	−0.0026	0.0850
FTP	0.0003	0.0000	0.2206	0.0984	1.0811	0.0427	−0.0023	0.0868
EIP	0.0091	0.0007	0.2474	0.1232	1.0966	0.2400	−0.0053	0.0650
GMVP	0.0433	0.0041	0.2053	0.3152*	1.2866	0.1766	0.0327	0.2635*
EWP	0.0057	0.0006	0.2077	0.1304	1.1078	0.0340	0.0032	0.1206
EWI	−0.0186	−0.0015	0.2254	0.0125	1.0099	0.0000	−0.0186	0.0125
VTP ($\eta = 1$)	0.0181	0.0017	0.2048	0.1929	1.1610	0.0486	0.0152	0.1787
VTP ($\eta = 2$)	0.0294	0.0027	0.2037	0.2492	1.2105	0.0611	0.0257	0.2312
VTP ($\eta = 3$)	0.0387	0.0035*	0.2044	0.2942*	1.2518	0.0762	0.0341	0.2717
VTP ($\eta = 4$)	0.0459	0.0041**	0.2063	0.3260**	1.2821	0.0916	0.0403	0.2993*

Notes: This table reports the out-of-sample annualized excess return (ER), Jensen's alpha (Alpha), annualized standard deviation (STD), Sharpe ratio (SR), Omega ratio (Omega), Turnover, annualized net excess return (NER), and net Sharpe ratio (NSR) for the examined portfolios. The significance levels of the Sharpe ratio for each portfolio, in comparison to the benchmark index, are derived using the method proposed by Ledoit and Wolf (2008). ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

values of volatility timing. Future research could explore incorporating other timing indicators into the tracking process and investigate ways to address the limitations of volatility timing.

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Data availability statement

Data can be downloaded from the Wind website and Kenneth French's website.

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Appendices

Appendix A. Proof of Proposition 1

The Lagrange multiplier method is applied to solve Model (4), and the first-order condition can be written as:

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{V}\mathbf{w} - \text{cov}(\mathbf{r}, \phi_{r_I}) - \lambda \mathbf{1} = \mathbf{0}, \quad (\text{A1})$$

$$\frac{\partial L}{\partial \lambda} = \mathbf{w}^T \mathbf{1} - 1 = 0, \quad (\text{A2})$$

where λ is the Lagrange multiplier. Based on Equation (A1), we can obtain

$$\mathbf{w} = \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi_{r_I}) + \lambda \mathbf{V}^{-1} \mathbf{1}. \quad (\text{A3})$$

By substituting Equation (A3) into Equation (A2), we can further get:

$$\lambda = \frac{1-a}{b}, \quad (\text{A4})$$

where $a = \mathbf{1}^T \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi_{r_I})$ and $b = \mathbf{1}^T \mathbf{V}^{-1} \mathbf{1}$.

Hence, the optimal tracking portfolio can be given as follows:

$$\mathbf{w}^* = a\mathbf{w}_V + (1-a)\mathbf{w}_G, \quad (\text{A5})$$

where $a = \mathbf{1}^T \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi_{r_I})$, $b = \mathbf{1}^T \mathbf{V}^{-1} \mathbf{1}$, $\mathbf{w}_V = \frac{1}{a} \mathbf{V}^{-1} \text{cov}(\mathbf{r}, \phi_{r_I})$, and $\mathbf{w}_G = \frac{1}{b} \mathbf{V}^{-1} \mathbf{1}$.

Appendix B. Benchmark portfolios

Benchmark index

The benchmark index is regarded as a buy-and-hold strategy that does not involve any parameter estimation and rebalance.

Equally weighted index (EWI)

The EWI assigns equal weights to the constituent stocks included in the index and is also regarded as a buy-and-hold strategy.

Traditional index tracking portfolio (ITP)

The ITP passively tracks the performance of the benchmark index (Rudolf et al. 1999), i.e.

$$\begin{cases} \min_{\mathbf{w}} \sigma^2(\mathbf{w}^T \mathbf{r} - r_I) \\ s.t. \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (\text{B1})$$

Factor tracking portfolio (FTP)

The FTP (Jiang et al. 2021) represents the case that the tracked benchmark contains an optimal tuning factor, i.e.

$$\begin{cases} \min_{\mathbf{w}, \eta} \sigma^2(\mathbf{w}^T \mathbf{r} - \eta r_I) \\ s.t. \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (\text{B2})$$

where η ensures that the tracking portfolio can flexibly track r_I .

Traditional enhanced indexation portfolio (EIP) (de Paulo et al. 2016):

$$\begin{cases} \min_{\mathbf{w}} \varrho \sigma^2(\mathbf{w}^T \mathbf{r} - r_I) - \varsigma(\mathbf{w}^T \boldsymbol{\mu} - \mu_B) \\ s.t. \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (\text{B3})$$

where $\boldsymbol{\mu}$ is the expected excess return vector of risky assets and μ_B is the expected excess return of the benchmark index.

Equally weighted portfolio (EWP)

The EWP allocates an equal weight to each asset (DeMiguel et al. 2009). Hence, $w_i^{EWP} = 1/n$ for the i th asset. Moreover, the

EWP does not involve any parameter estimation and optimization techniques.

Volatility timing portfolio (VTP) (Kirby and Ost diek 2012)

The VTP contains a similar timing mechanism to the volatility timing benchmark and is also regarded as a benchmark. The weight of each asset is:

$$w_i^{VTP} = \left(\frac{1}{\sigma_i^2} \right)^\eta / \sum_{i=1}^n \left(\frac{1}{\sigma_i^2} \right)^\eta, \quad (\text{B4})$$

where $\eta \in \{1, 2, 3, 4\}$.

Global minimum variance portfolio (GMVP)

The GMVP is an active portfolio that aims to minimize the variance of portfolio returns (Jiang et al. 2019), i.e.

$$\begin{cases} \min_{\mathbf{w}} \mathbf{w}^T \mathbf{V} \mathbf{w} \\ s.t. \mathbf{w}^T \mathbf{1} = 1 \end{cases} \quad (\text{B5})$$

where $\mathbf{V}$ is the covariance matrix of risky asset returns.

Appendix C. Performance metrics

We assess the out-of-sample performance of a portfolio using the following metrics.

Excess return (ER):

$$ER = \frac{1}{K-L} \sum_{t=L+1}^K (r_{P,t} - r_{I,t}), \quad (\text{C1})$$

where $r_{P,t}$ is the out-of-sample return of the portfolio at time t .

Jensen's alpha (Alpha):

$$\text{Alpha} = E[r_{P,t}] - \beta E[r_{I,t}], \quad (\text{C2})$$

where $\beta = \text{cov}(r_{P,t}, r_{I,t}) / \sigma_I^2$ represents the systematic risk of the portfolio relative to the benchmark index. $r_{P,t}$ and $r_{I,t}$ are the returns in excess of the risk-free rate.

Standard deviation (STD):

$$\hat{\sigma}^{\text{Out}} = \sqrt{\frac{1}{K-L} \sum_{t=L+1}^K (r_{P,t} - \bar{\mu}^{\text{Out}})^2}, \quad (\text{C3})$$

where $\hat{\mu}^{\text{Out}} = \frac{1}{K-L} \sum_{t=L+1}^K r_{P,t}$.

Sharpe ratio (SR):

$$SR = \frac{\hat{\mu}^{\text{Out}}}{\hat{\sigma}^{\text{Out}}}. \quad (\text{C4})$$

Omega ratio (Omega):

$$\text{Omega} = \frac{E[\max(0, r_{P,t} - \tau)]}{E[\max(0, \tau - r_{P,t})]}, \quad (\text{C5})$$

where $\tau = 0$ is a threshold distinguishing between gains and losses of the portfolio.

Turnover (Kirby and Ost diek 2012):

$$\pi = \frac{1}{K-L} \sum_{t=L}^K \sum_{i=1}^n |w_{i,t+1} - w_{i,t}|, \quad (\text{C6})$$

where $w_{i,t+1}$ is the i -th asset's weight of the portfolio at time $t+1$, and $w_{i,t}$ is the i -th asset's weight of the portfolio before rebalancing at time $t+1$.

Net excess return (NER) and net Sharpe ratio (NSR): Both are computed based on the net out-of-sample returns of the portfolio, which is given as follows:

$$r_{N,t} = (1 + r_{P,t})(1 - \pi_t c) - 1, \quad (\text{C7})$$

where $c = 50\text{bps}$ is the transaction cost (Kirby and Ost diek 2012).

Appendix D. Lists of sector indexes

Table D1. The sectors for each dataset.

The name of each sector	Code	Abbreviation
Panel A: The US sectors of the S&P 500 dataset		
S&P 500 Consumer Discretionary Sector Index	S5COND.SPI	Condc
S&P 500 Consumer Staples Sector Index	S5CONS.SPI	Consp
S&P 500 Health Care Sector Index	S5HLTH.SPI	Helcr
S&P 500 Industrials Sector Index	S5INDU.SPI	Indst
S&P 500 Information Technology	S5INFT.SPI	Inftc
S&P 500 Materials Sector Index	S5MATR.SPI	Mteri
S&P 500 Communications Services Sector Index	S5TELS.SPI	Comsr
S&P 500 Utilities Sector Index	S5UTIL.SPI	Utilities
S&P 500 Financials Sector Index	SPF.SPI	Fincl
S&P 500 Energy Sector Index	SPN.SPI	Energy
S&P 500 Real Estate Sector Index	S5RLST.SPI	Reaet
Panel B: The Chinese sectors of the CSI 300 dataset		
CSI 300 Energy Index	000908.SS	Energy
CSI 300 Materials Index	000909.SS	Mteri
CSI 300 Industrials Index	000910.SS	Indst
CSI 300 Consumer Discretionary	000911.SS	Condc
CSI 300 Consumer Staples Index	000912.SS	Consp
CSI 300 Health Care Index	000913.SS	Helcr
CSI 300 Financials Index	000914.SS	Fincl
CSI 300 Information Technology	000915.SS	Inftc
CSI 300 Telecommunication Services	000916.SS	Telcm
CSI 300 Utilities Index	000917.SS	Utilities